PENGOLAHAN CITRA & VISI KOMPUTER

Human Vision & Enhancement

(sumber: R.C. Gonzalez & R.E. Woods, Digital Image Processing, 4th edition)

Handayani Tjandrasa

Struktur Mata Manusia

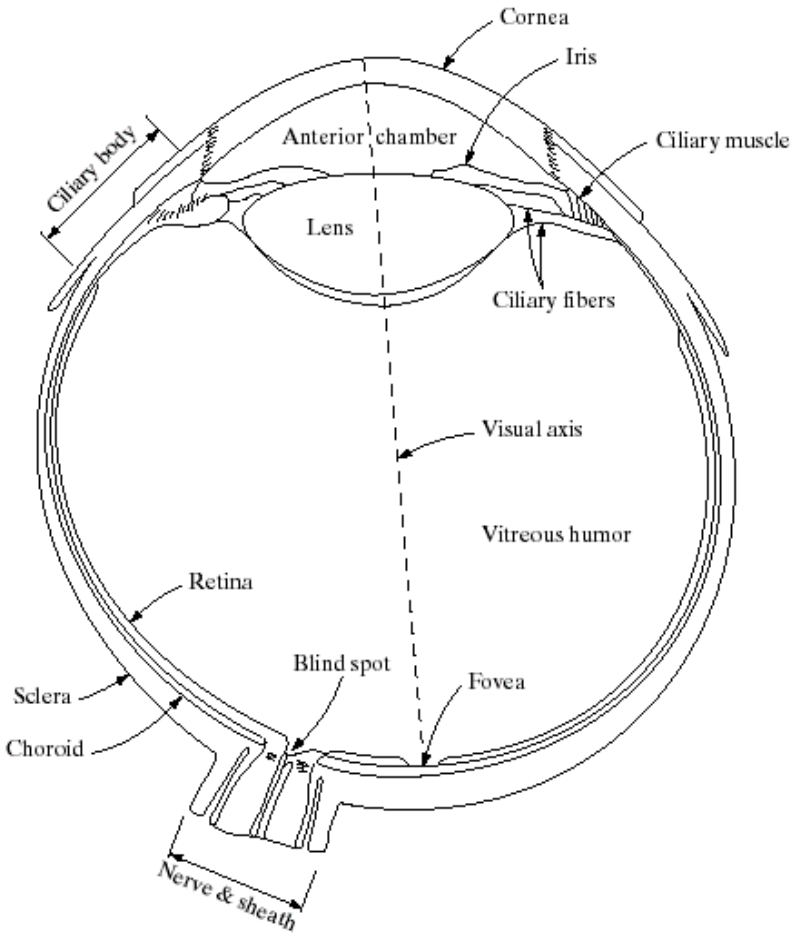


FIGURE 2.1
Simplified diagram of a cross section of the human eye.

- Diameter rata=20mm
- Cornea: transparan, melindungi bag. anterior
- Sclera: melindungi sisanya
- Choroid dibawah sclera berisi pembuluh darah, terpigmentasi
- Lensa: mencegah infrared & ultraviolet berlebihan
- Membran terdalam, dimana cahaya objek dicitrakan

Distribusi *Rods & Cones* pada Retina

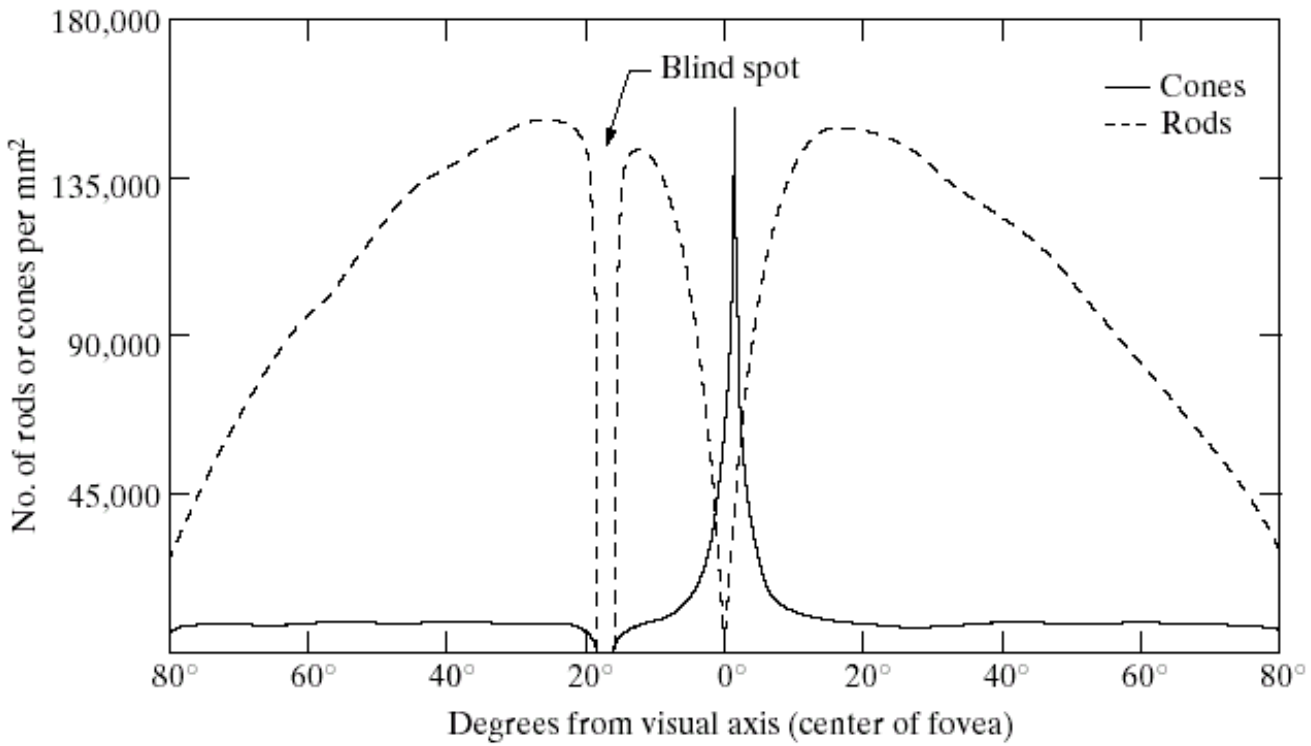


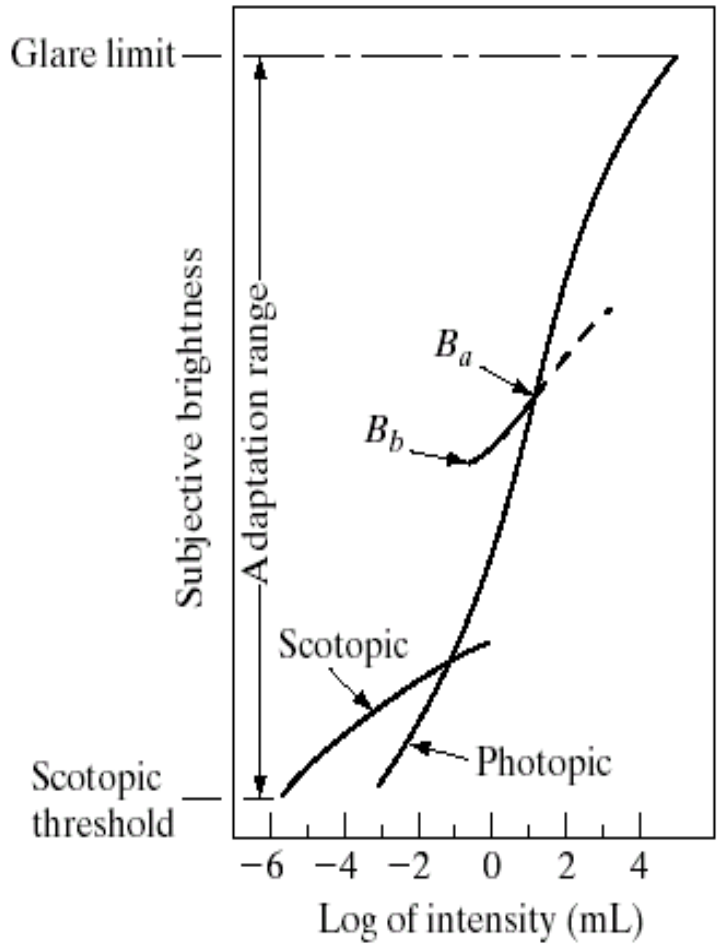
FIGURE 2.2
Distribution of rods and cones in the retina.

#cones=6-7 juta, color, masing2 terhubung ke 1 ujung saraf, detil, photopic (bright-light vision)

#rods=75-150 juta, beberapa terhubung ke 1 ujung saraf, kurang detil, scotopic (dim-light vision)

Adaptasi *brightness*

FIGURE 2.4
Range of subjective brightness sensations showing a particular adaptation level.



Diskriminasi *brightness*

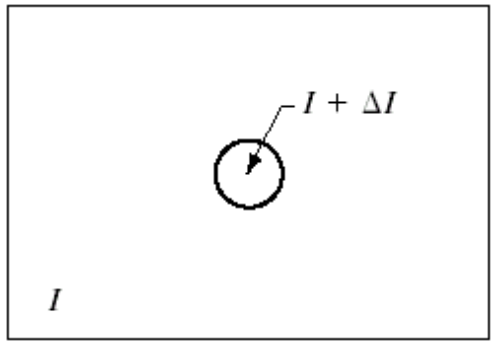


FIGURE 2.5 Basic experimental setup used to characterize brightness discrimination.

Weber ratio=ΔI/I

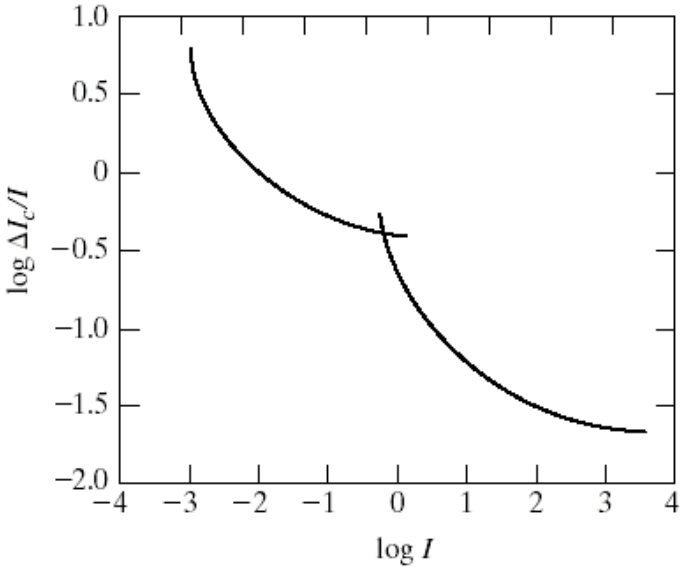
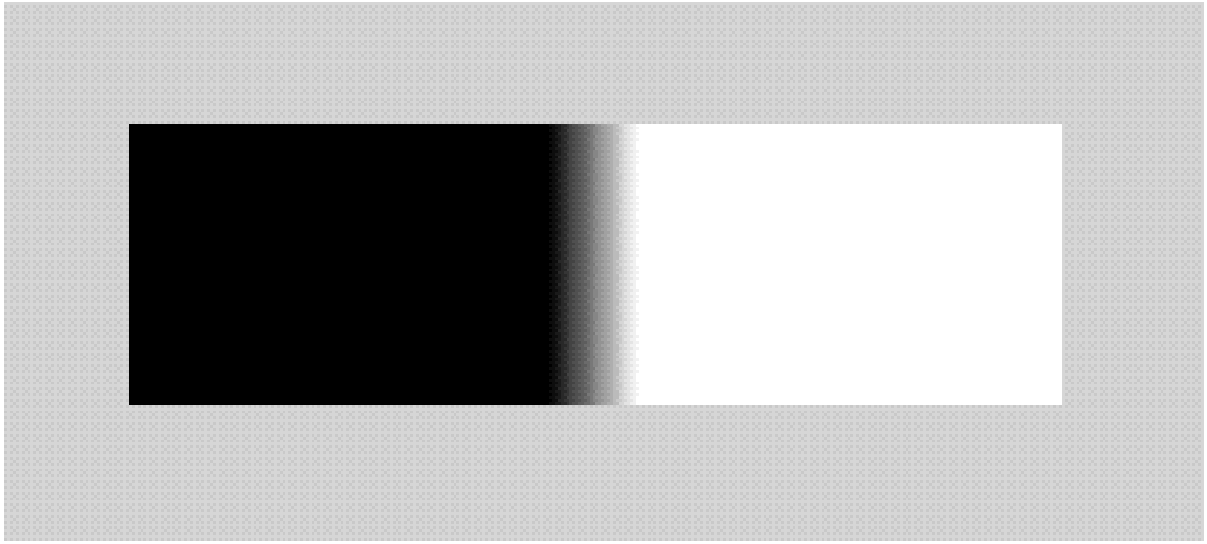
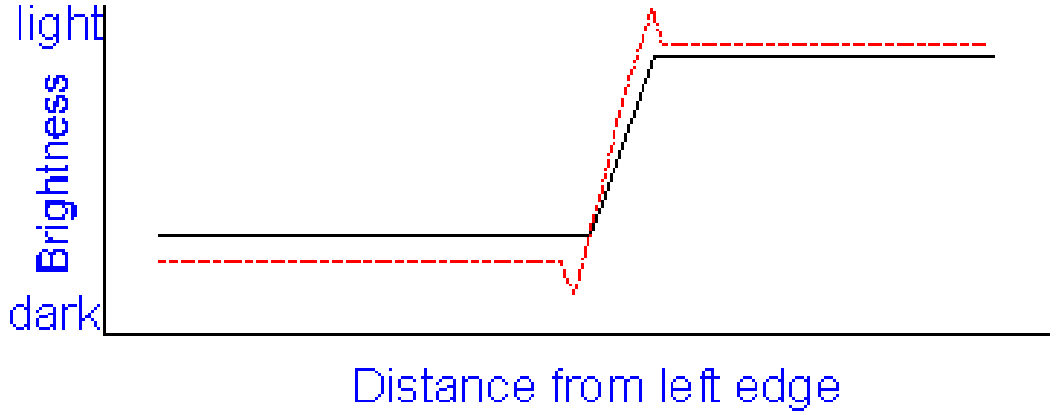


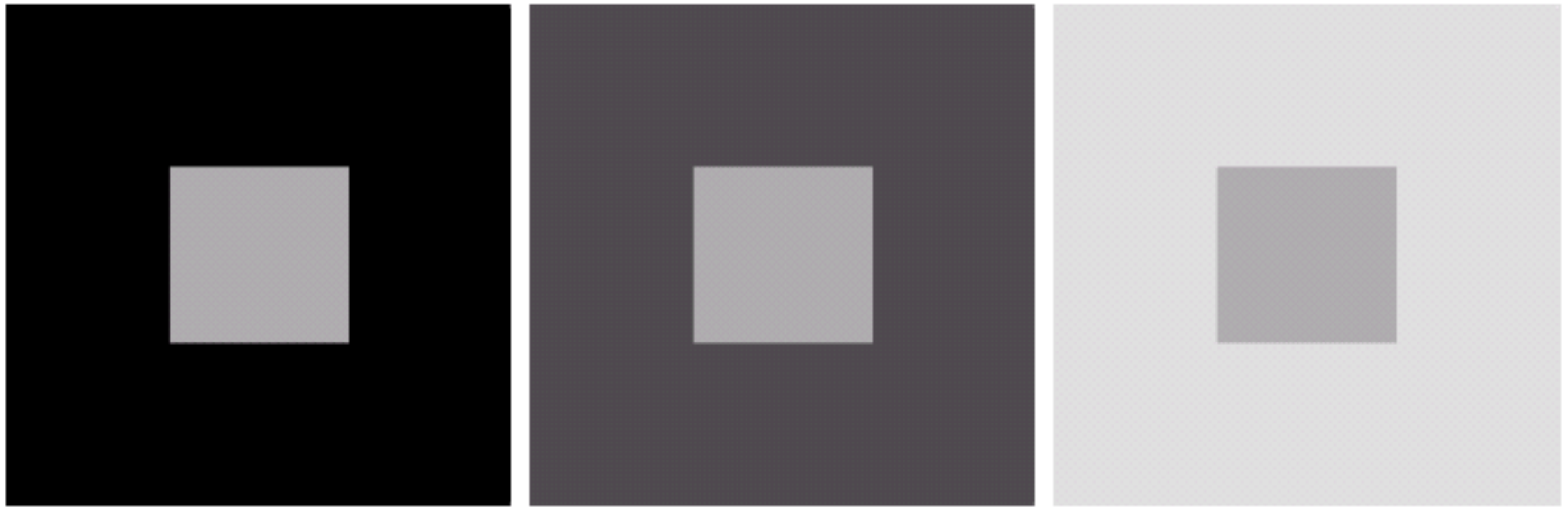
FIGURE 2.6 Typical Weber ratio as a function of intensity.

Match bands





Kontras simultan



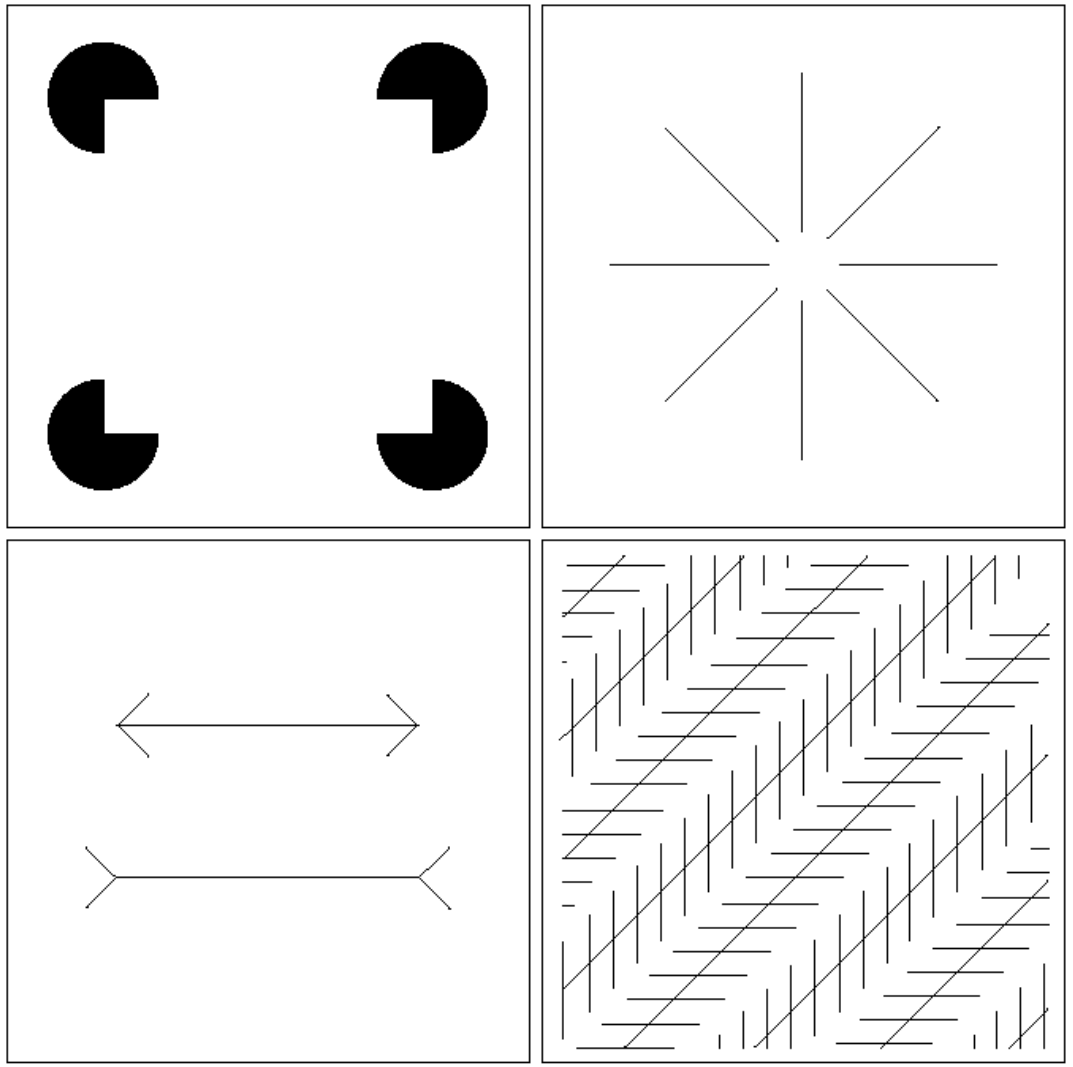
a b c

FIGURE 2.8 Examples of simultaneous contrast. All the inner squares have the same intensity, but they appear progressively darker as the background becomes lighter.

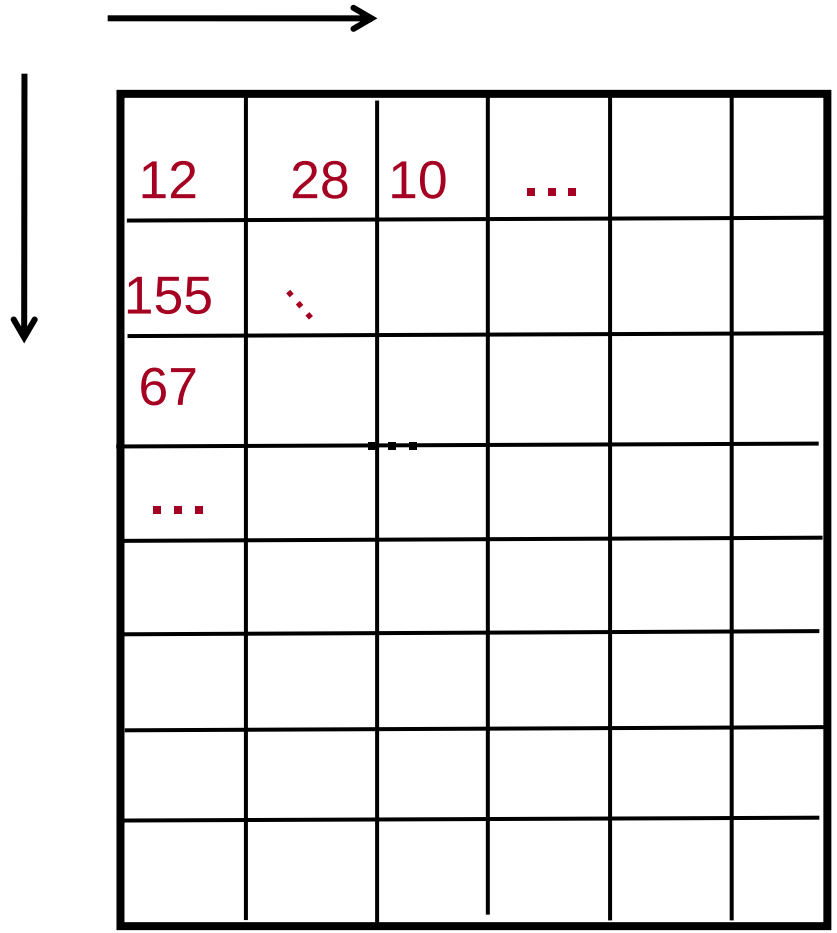
Ilusi Optikal

a b
c d

FIGURE 2.9 Some well-known optical illusions.



Definisi Citra Digital



Representasi Matrik

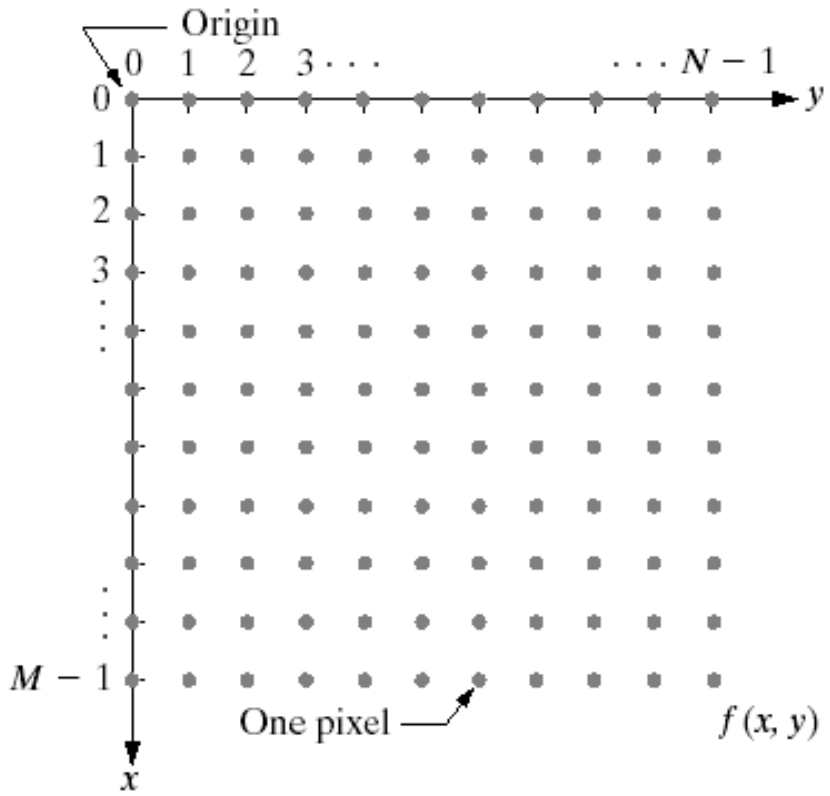


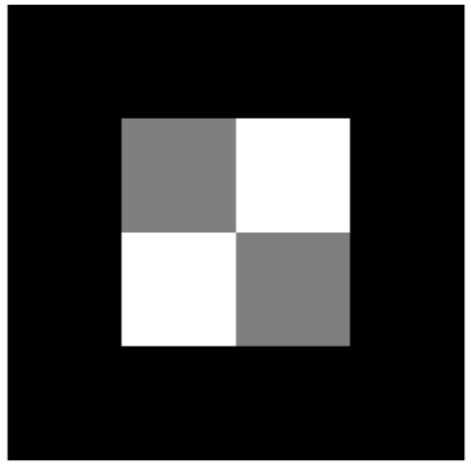
FIGURE 2.18
Coordinate convention used in this book to represent digital images.

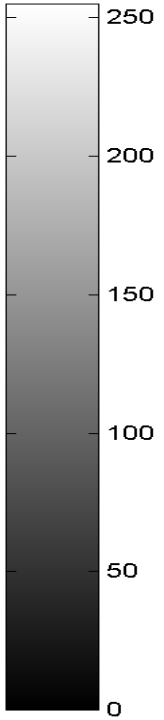
Citra Digital

Image matrix

0	0	0	0
0	127	255	0
0	255	127	0
0	0	0	0

Screen

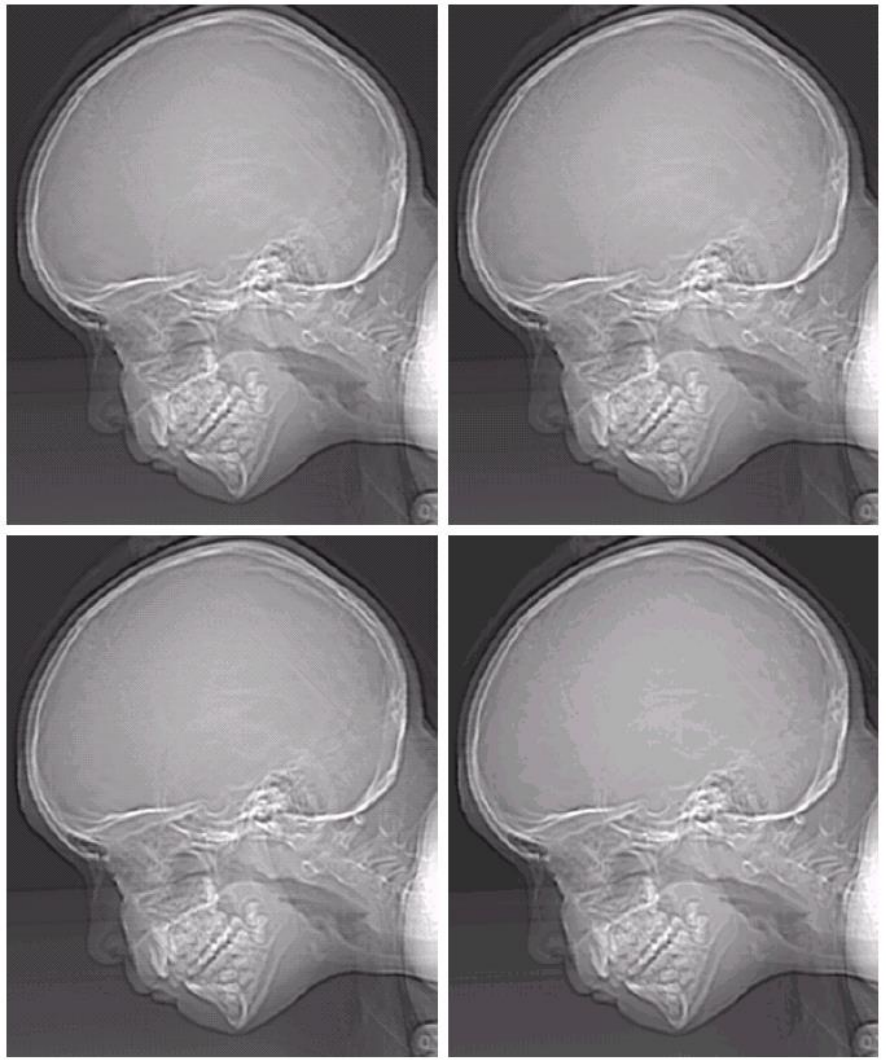




Efek Penurunan Resolusi



Efek Penurunan Resolusi



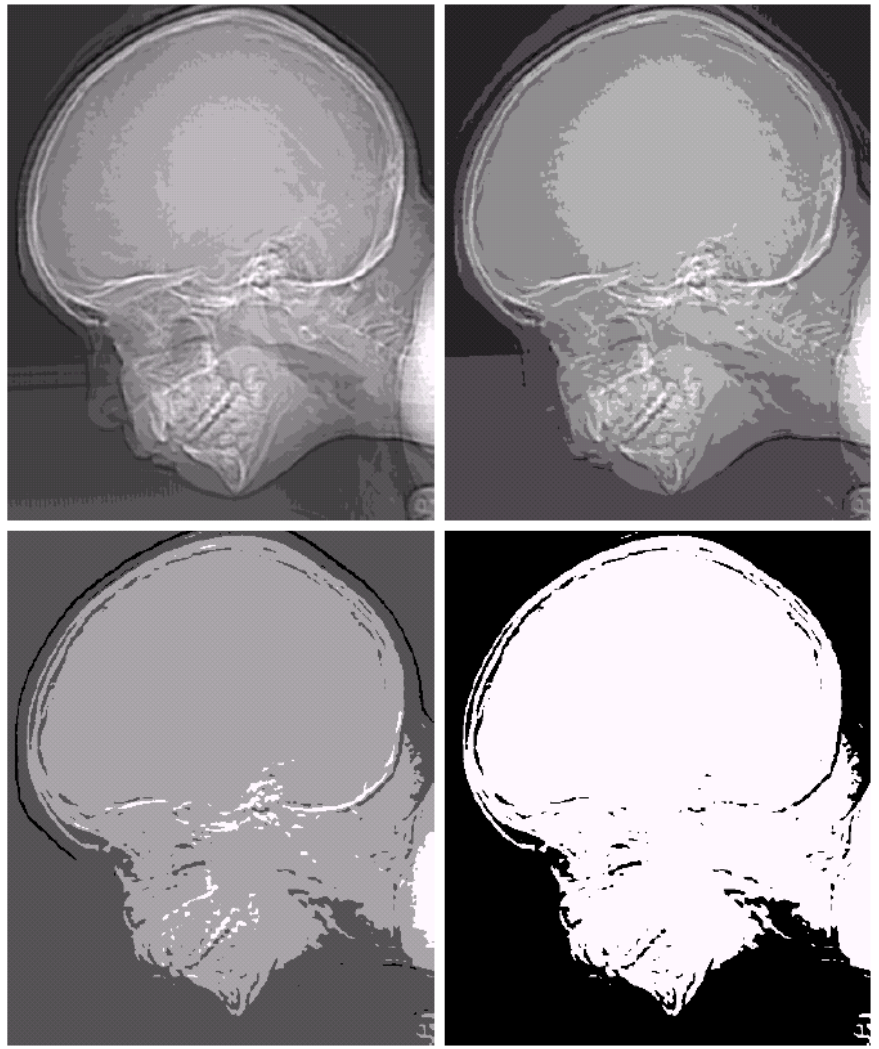
a b
c d

FIGURE 2.21
(a) 452 × 374, 256-level image. (b)–(d) Image displayed in 128, 64, and 32 gray levels, while keeping the spatial resolution constant.

Efek Penurunan Resolusi

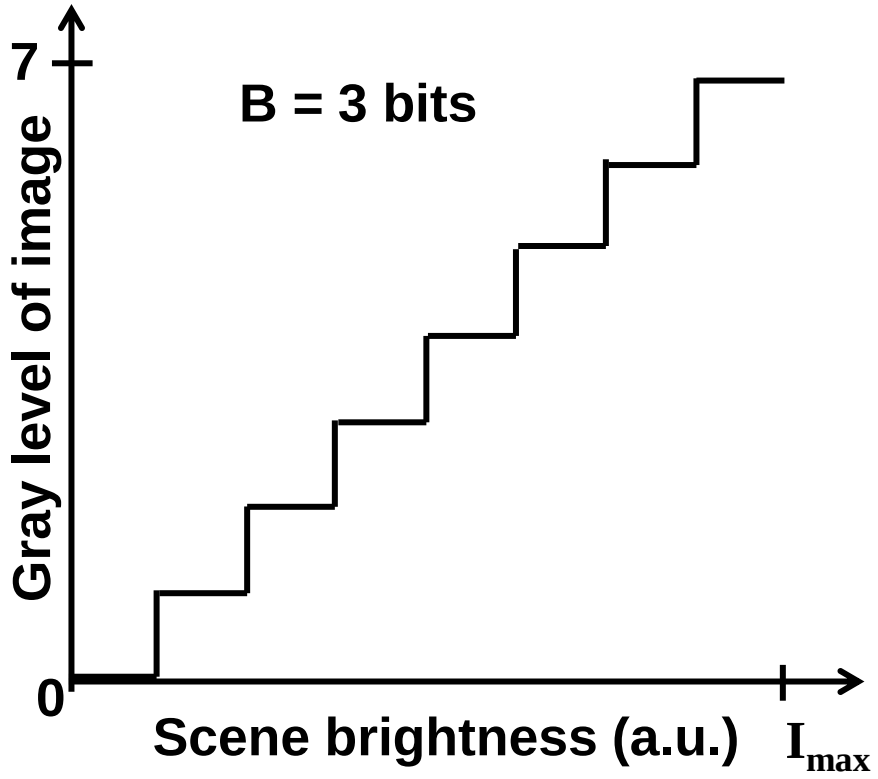
e f
g h

FIGURE 2.21
(Continued)
(e)–(h) Image displayed in 16, 8, 4, and 2 gray levels. (Original courtesy of Dr. David R. Pickens, Department of Radiology & Radiological Sciences, Vanderbilt University Medical Center.)

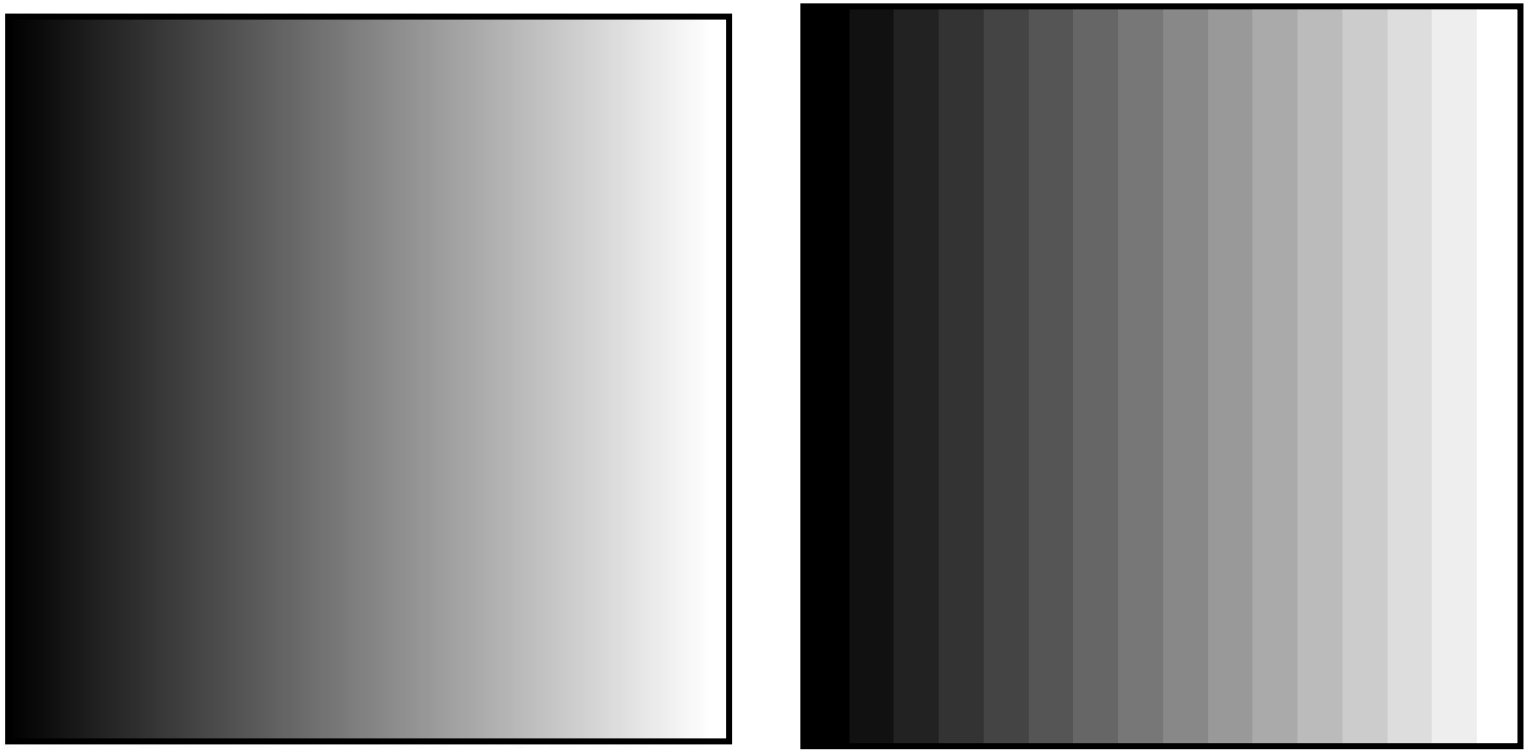


Resolusi *Brightness*

- **Pada kuantisasi, *brightness* di titik yg disampel pd citra kontinyu diganti dg harga integer**



Efek Digitasi



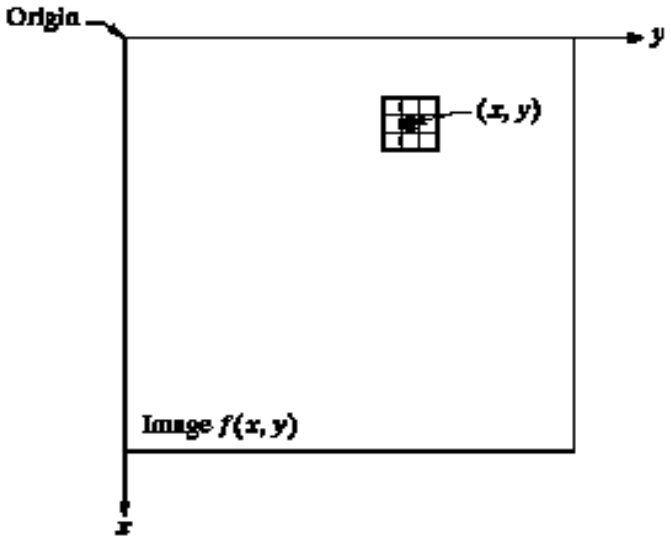
Perbaikan Citra (Enhancement)

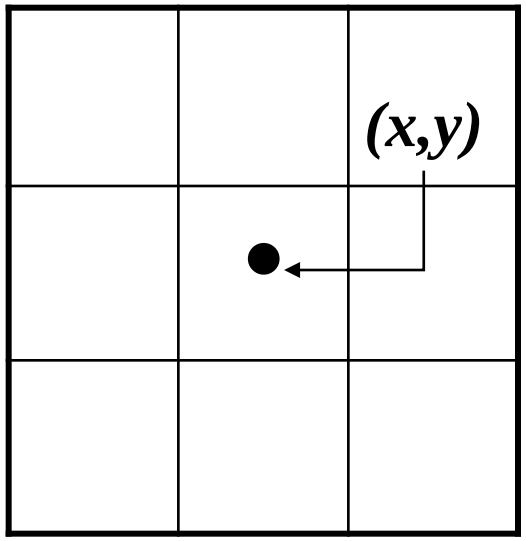
- Memproses citra supaya hasilnya sesuai dengan keinginan operator
- Metode mana yang cocok tergantung dari citra dan aplikasinya

Domain pemrosesan

- **Spatial domain: manipulasi langsung piksel-piksel citra**
- **Frequency domain: manipulasi pada transformasi Fourier citra**
- **Metode kombinasi dari kedua cara diatas**

Proses pada Spatial domain:
g(x,y) = T[f(x,y)]



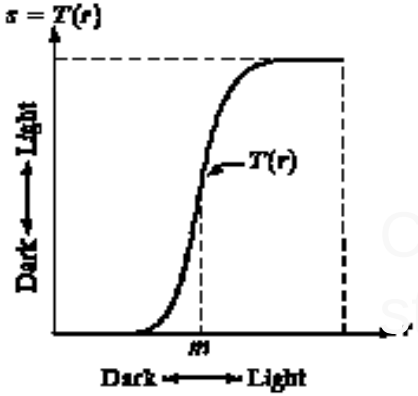


Point Processing

s = T(r)

r = gray level ***f(x,y)***

s = gray level ***g(x,y)***



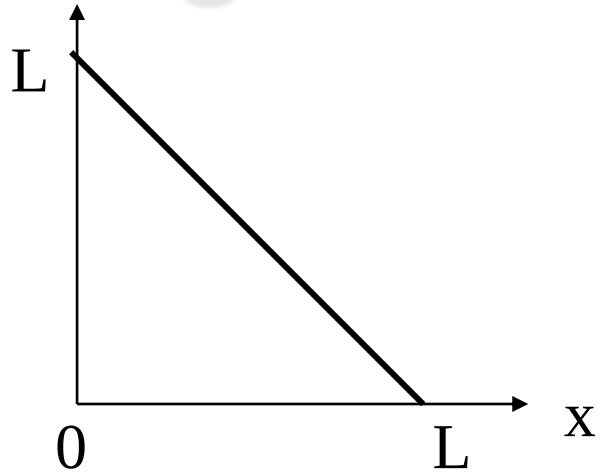
Filter

Sharpening &Smoothing

Contrast stretching

Proses Negatif

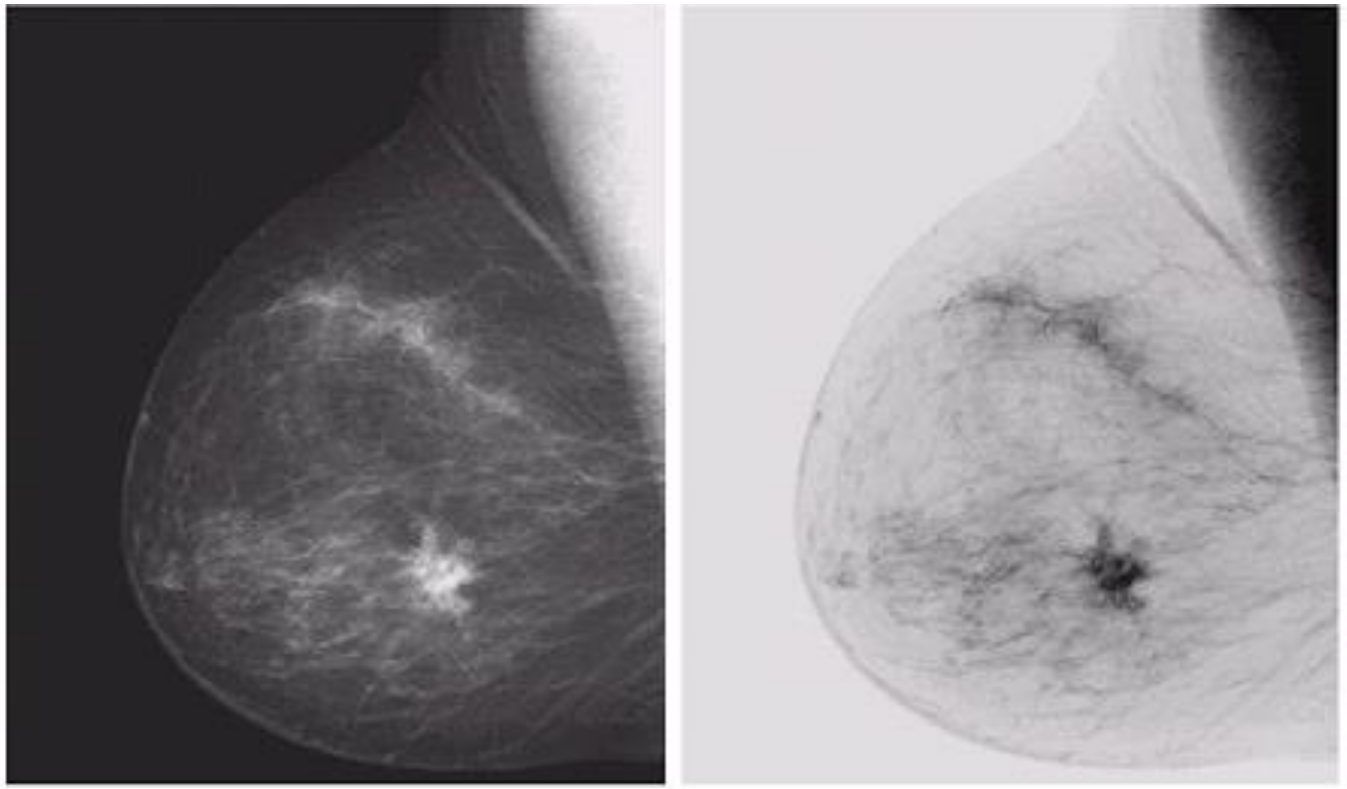
$$y = L - x$$





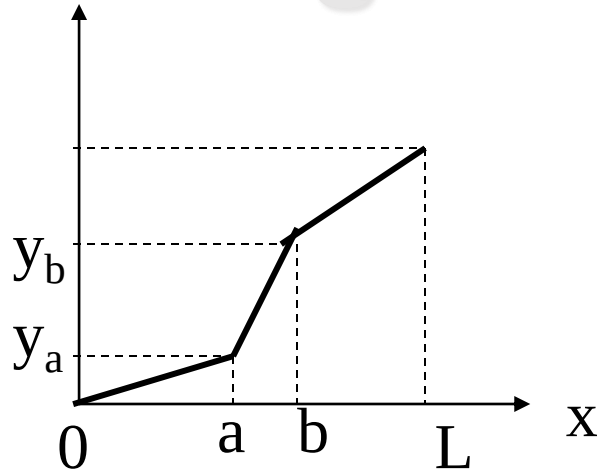


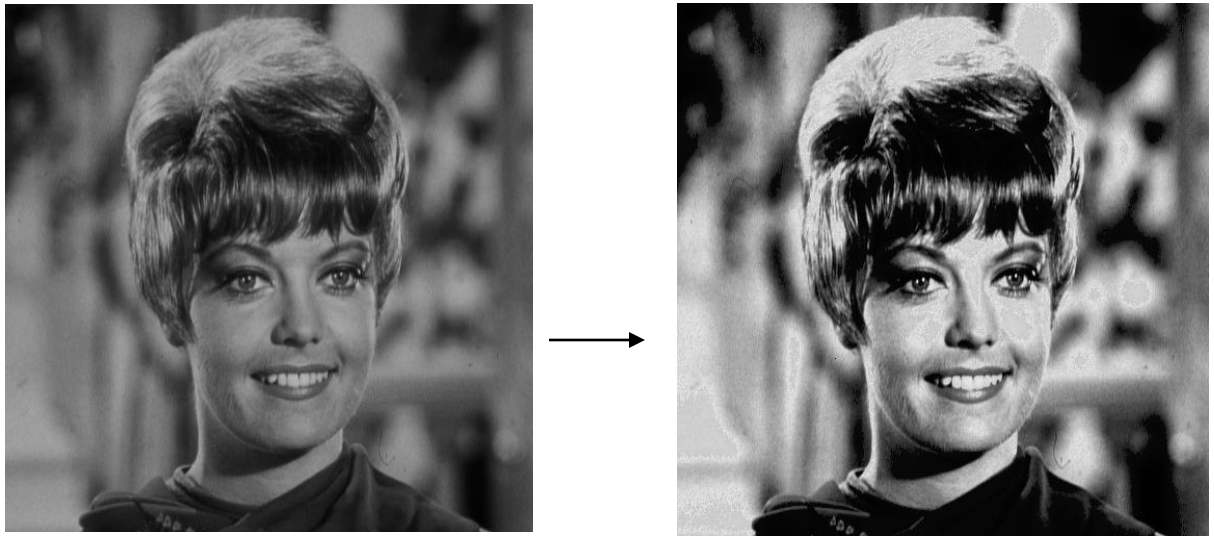
Contoh: Citra negatif memberikan gambar yang lebih jelas untuk analisis kelainan



Contrast Stretching

$$y = \begin{cases} \alpha x & 0 \le x < a \\ \beta(x-a) + y_a & a \le x < b \\ \gamma(x-b) + y_b & b \le x < L \end{cases}$$

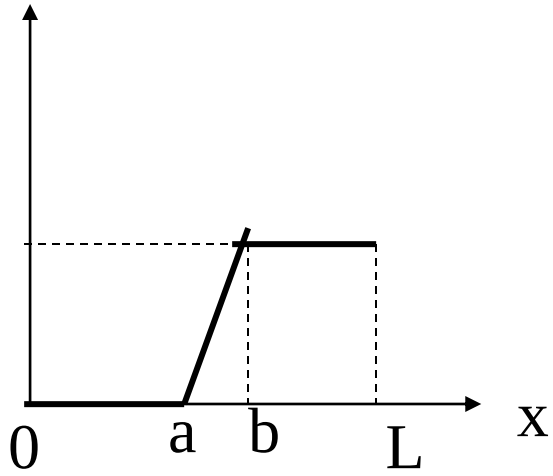




$a = 50, b = 150, \alpha = 0.2, \beta = 2, \gamma = 1, y_a = 30, y_b = 200$

Clipping

$$y = \begin{cases} 0 & 0 \le x < a \\ \beta(x-a) & a \le x < b \\ \beta(b-a) & b \le x < L \end{cases}$$



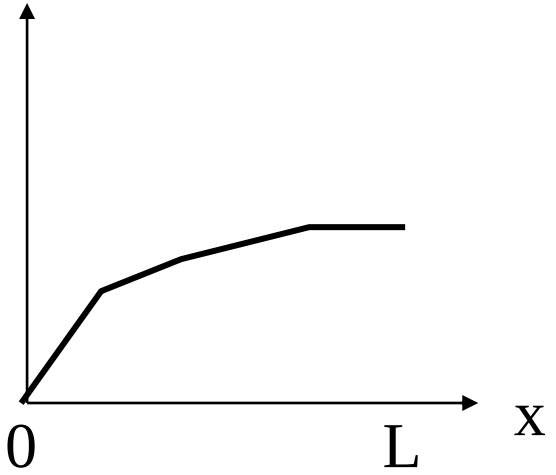




$a = 50, b = 150, \beta = 2$

Logaritmik

$$y = c\log_{10}(1+x)$$

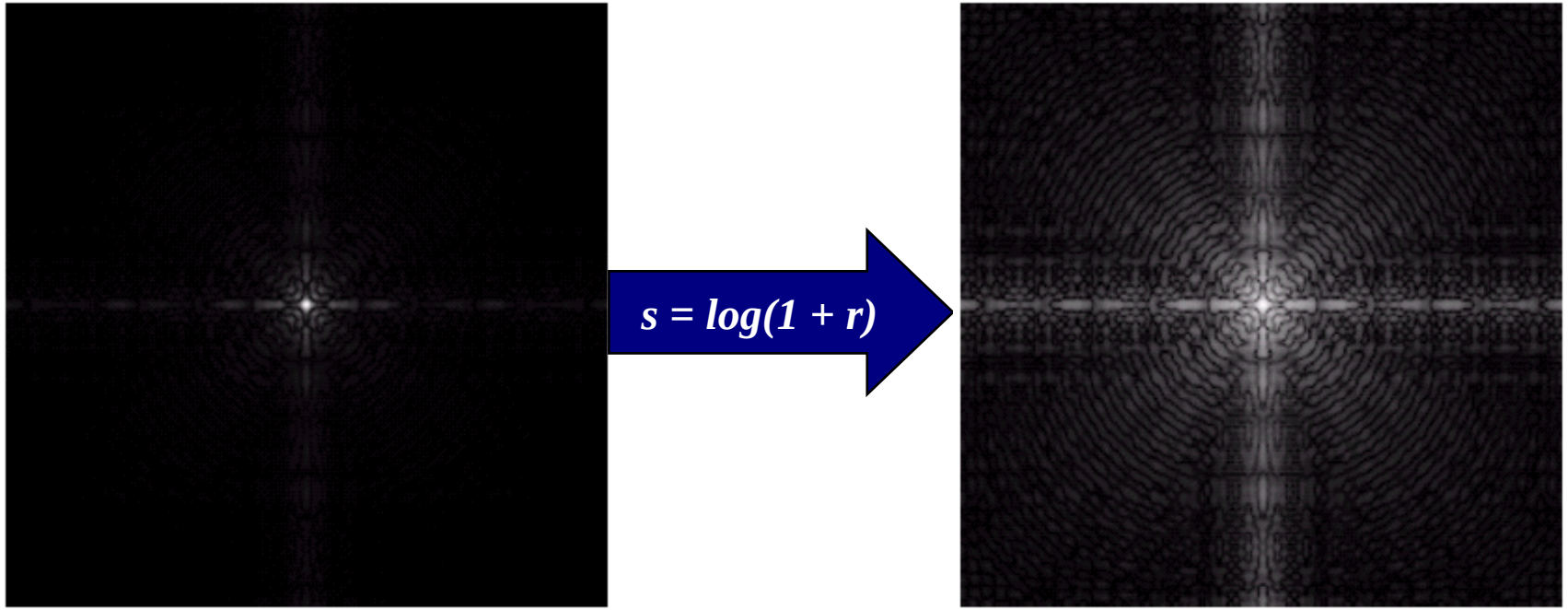






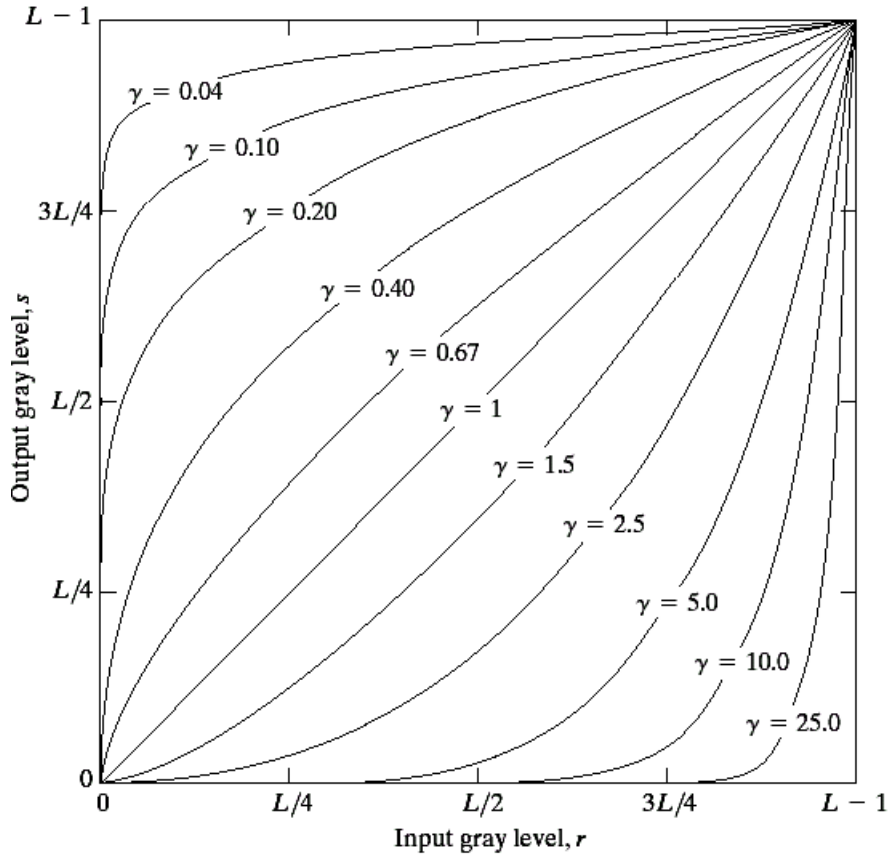
c=100

Contoh Transformasi Logaritmik



Transformasi Power Law

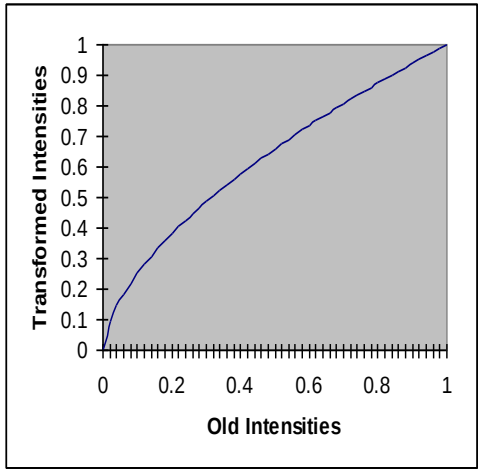
$$s = c * r^{\gamma}$$

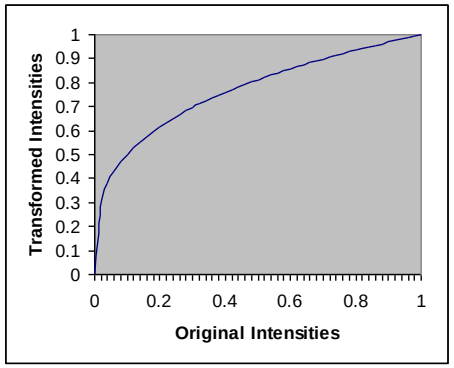


Contoh Transformasi Power Law

Parameter 0.6 dan 0.3





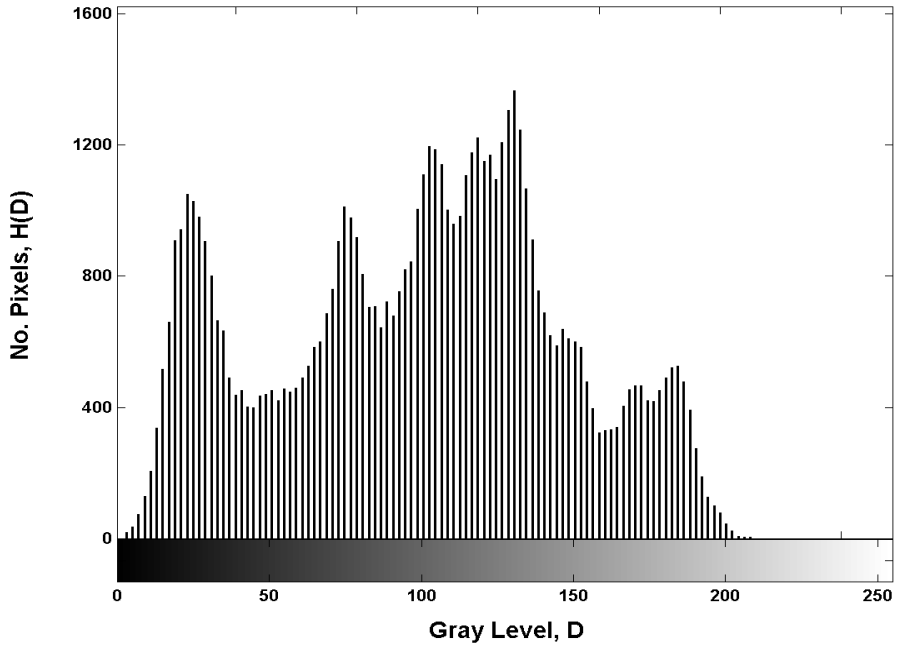






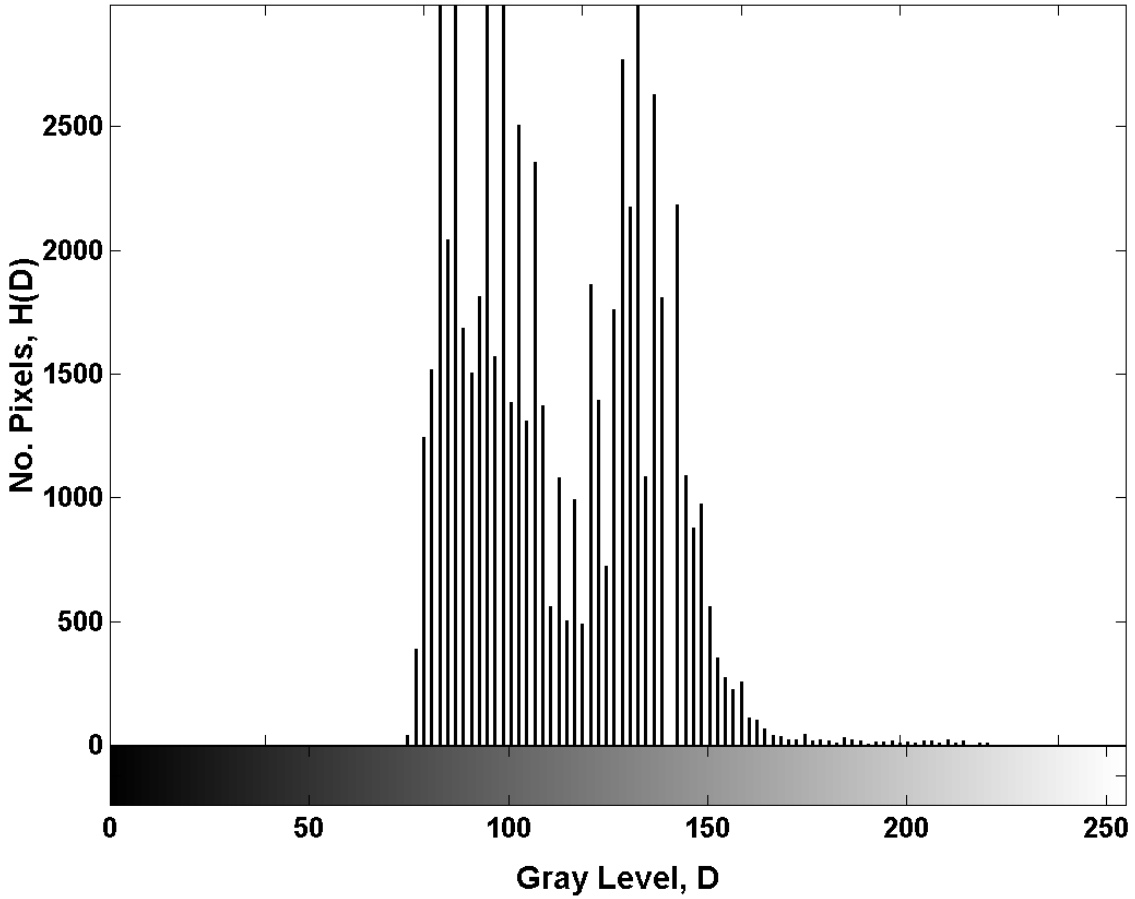
Definisi Histogram





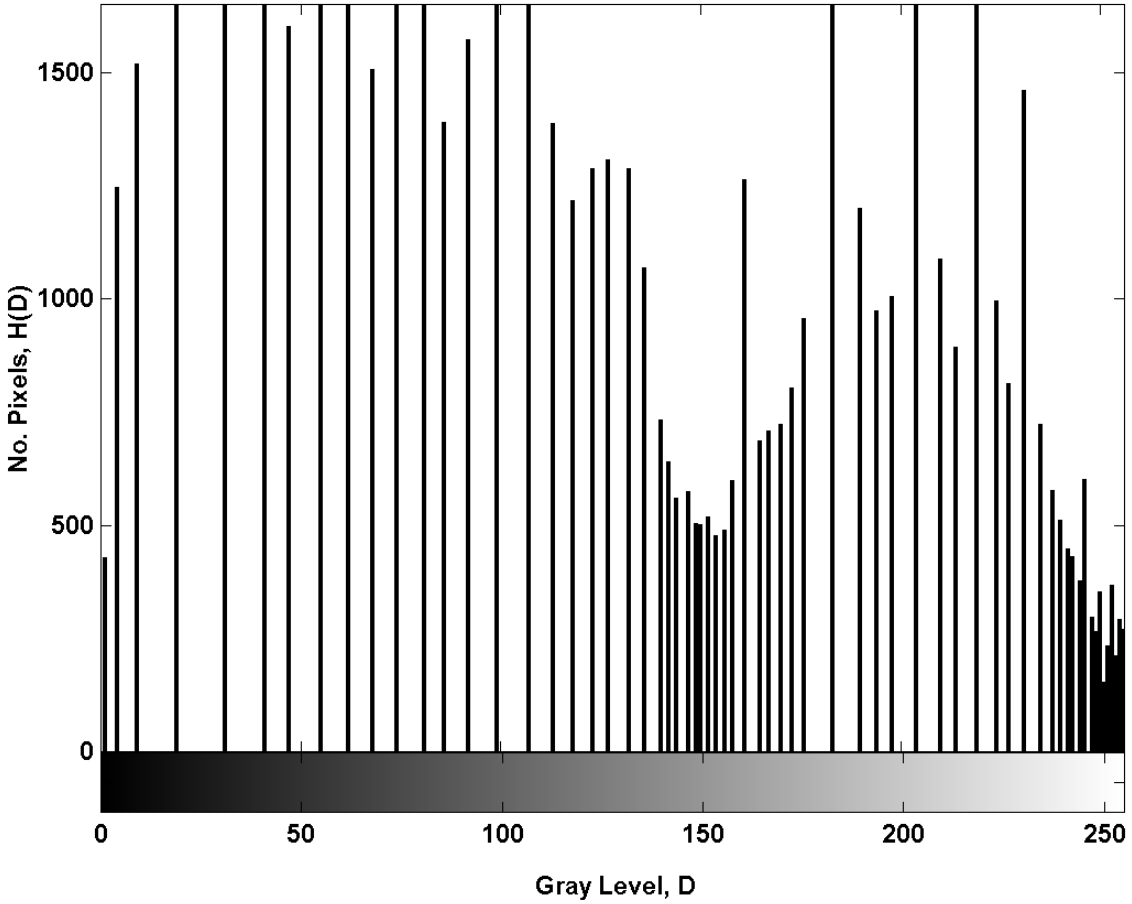
Histogram




2500
2000
1500
1000
500
0
No. Pixels, H(D)
0
50
100
150
200
250
Gray Level, D

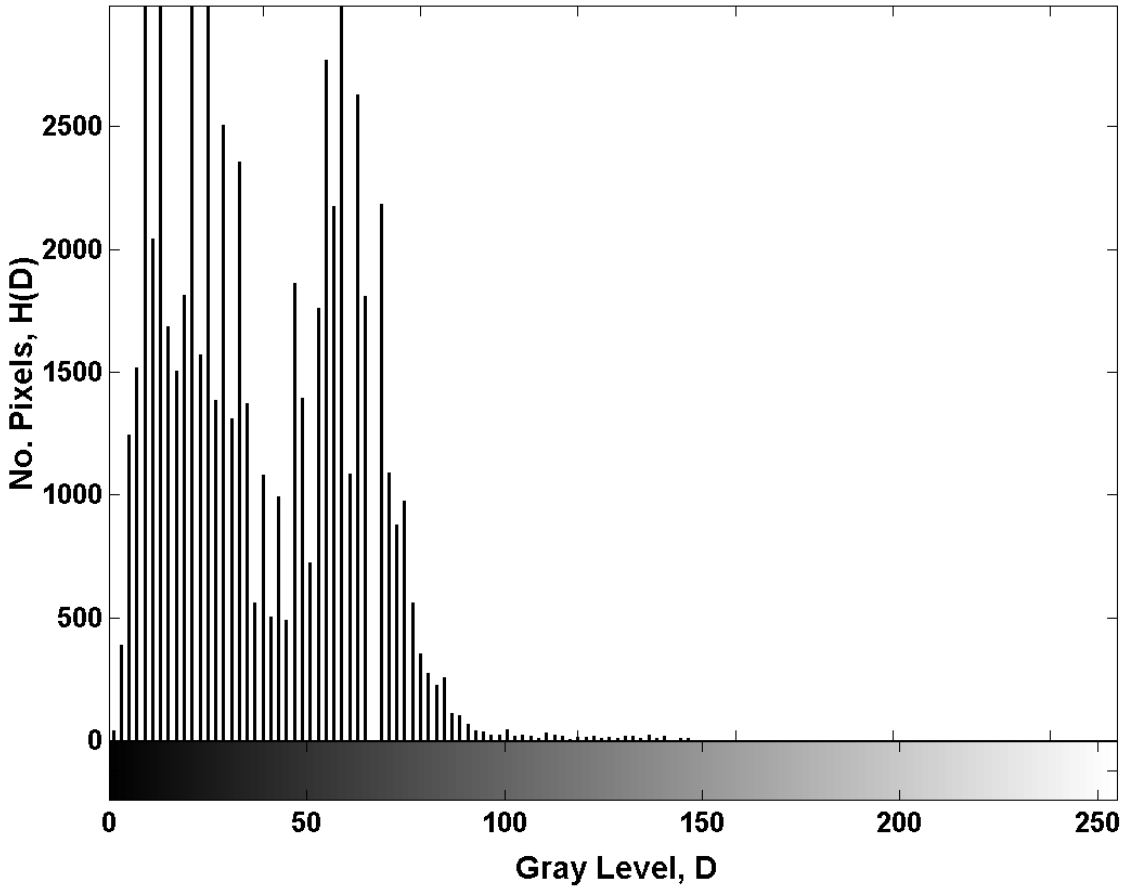
Histogram dengan Kontras lebih tinggi



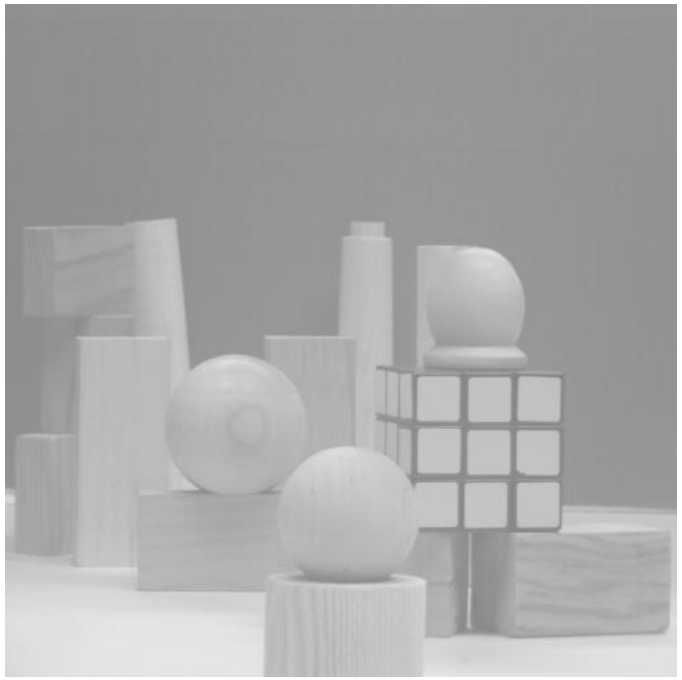


Gambar Gelap

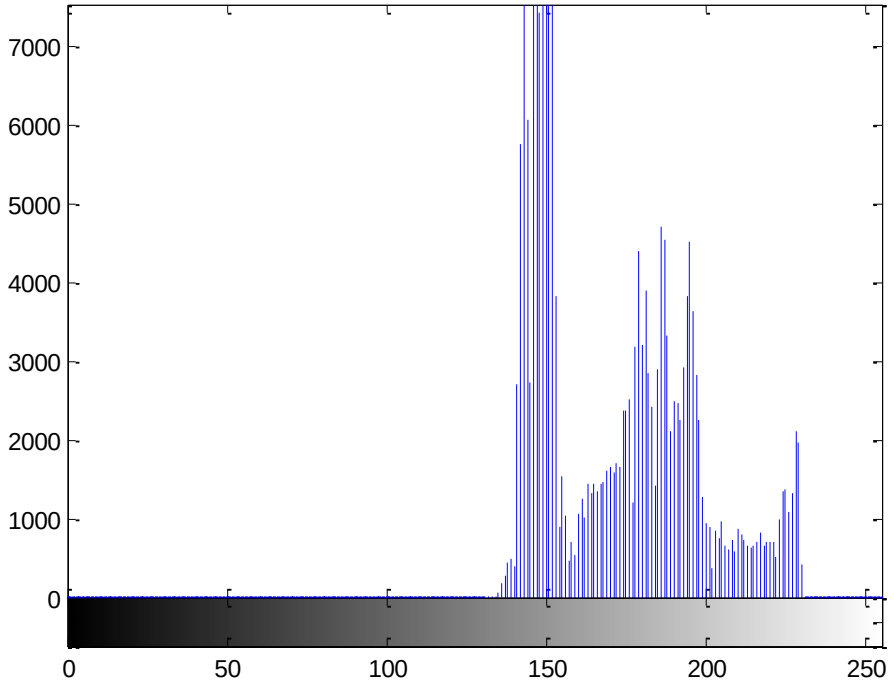




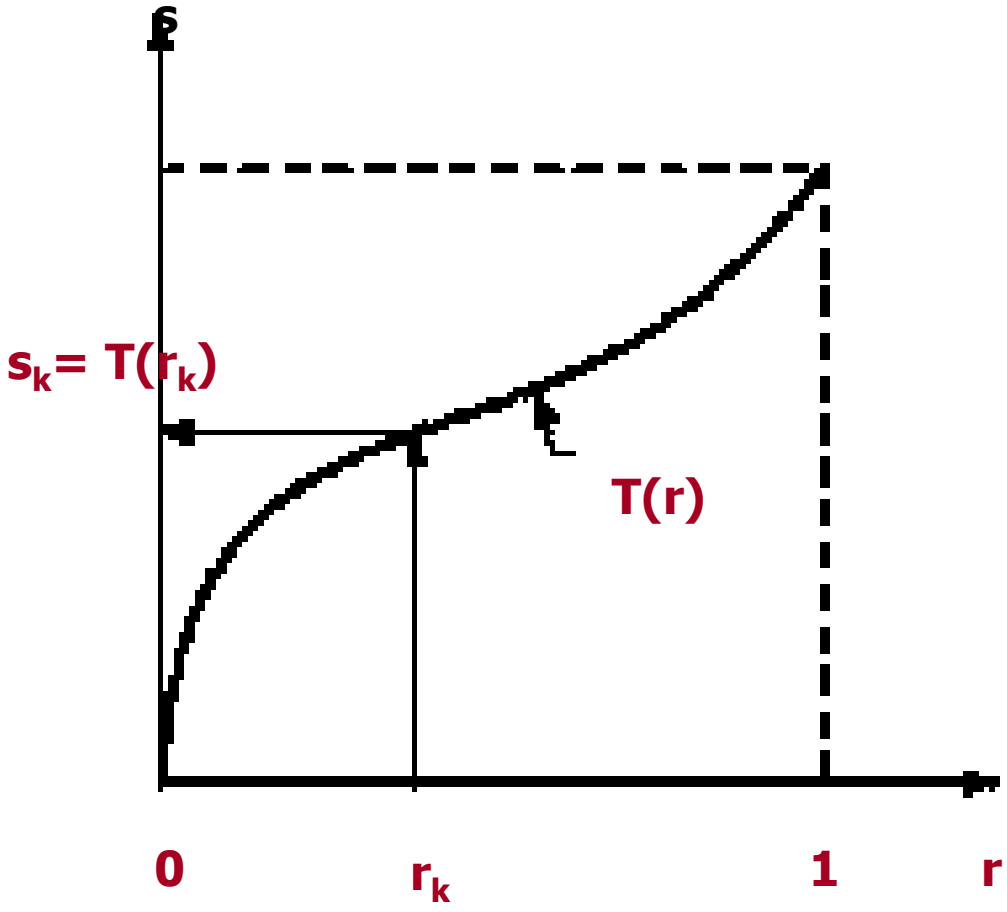
Gambar terekspos lebih



Over-exposed image



Transformasi Histogram



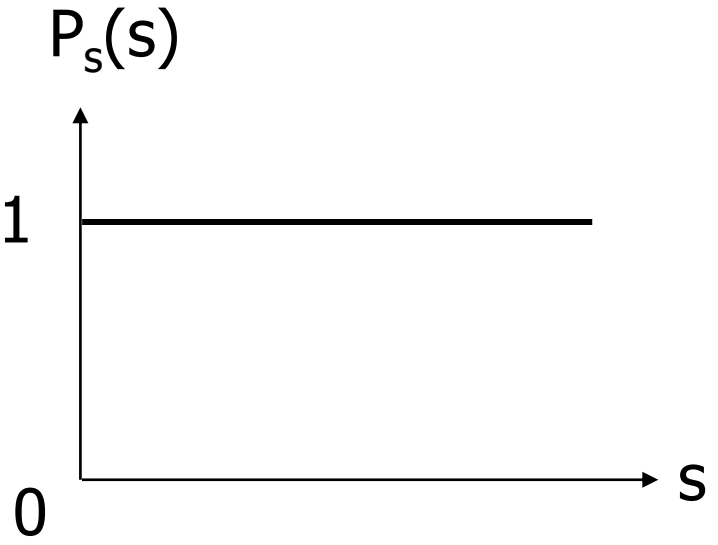
$$s = T(r)$$

- $0 \leq r \leq 1$
- T(r) berharga tunggal dan naik secara monoton pada $0 \leq r \leq 1$
- $0 \leq T(r) \leq 1$ untuk $0 \leq r \leq 1$

Histogram Equalization

$$p_s(s) = p_r(r)\left|\frac{dr}{ds}\right|$$

$$s = T(r) = \int_0^r p_r(w)dw$$



Dalam bentuk Diskrit

$$p_r(r_k) = \frac{n_k}{n} \qquad \textbf{where } k = 0, 1, ..., L\text{-}1$$

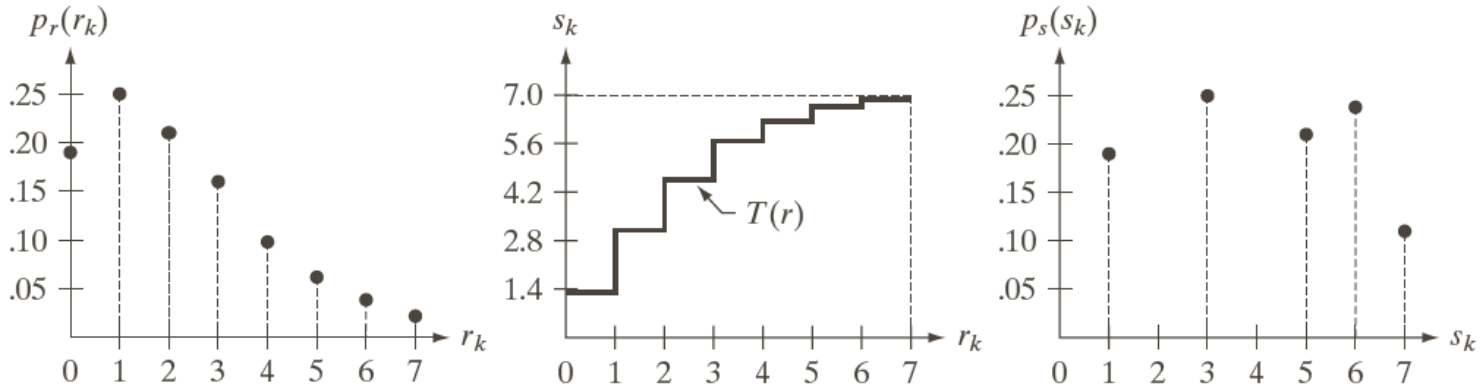
$$s_k = T(r_k) = \sum_{j=0}^{k} p_r(r_j)$$

$$= \sum_{j=0}^{k} \frac{n_j}{n} \qquad where\ k = 0, 1, ..., L\text{-}1$$

Contoh

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

TABLE 3.1
Intensity distribution and histogram values for a 3-bit, 64 × 64 digital image.



a b c

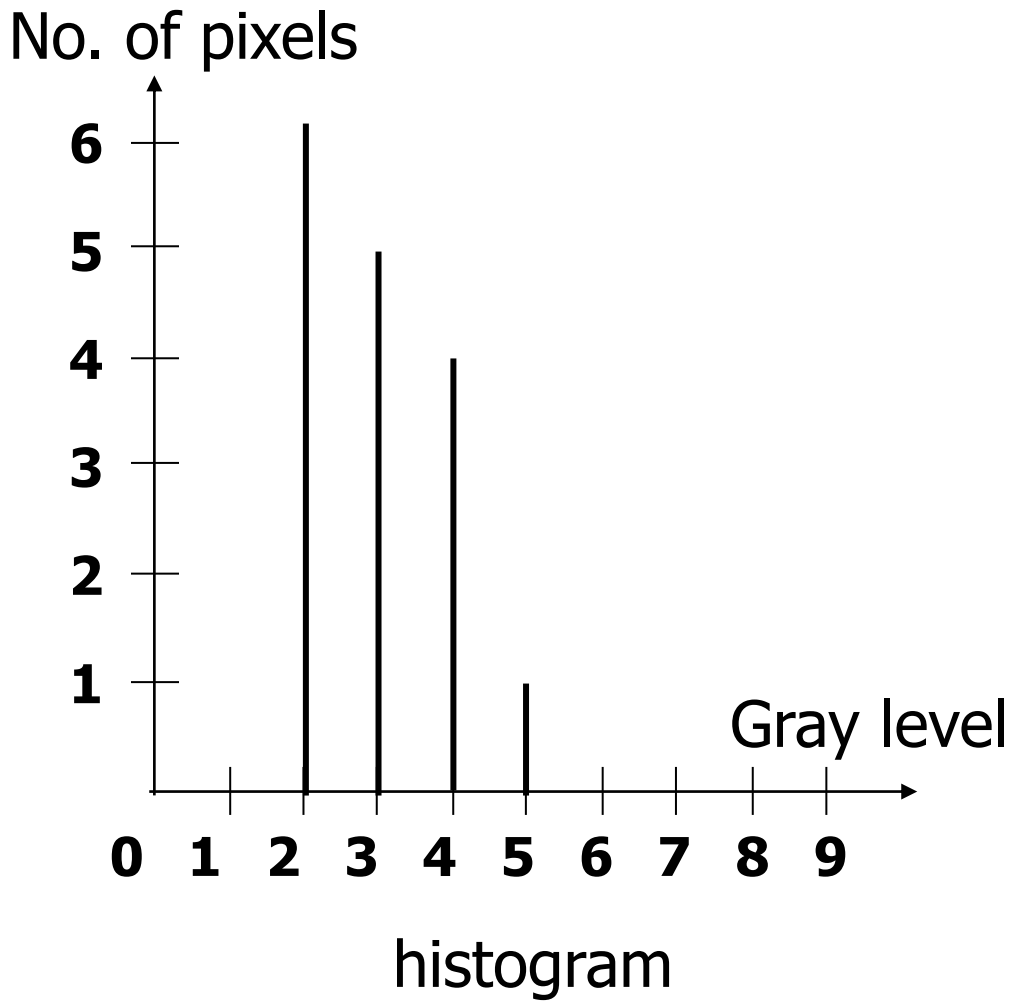
FIGURE 3.19 Illustration of histogram equalization of a 3-bit (8 intensity levels) image. (a) Original histogram. (b) Transformation function. (c) Equalized histogram.

Contoh

2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4

4x4 image

Gray scale = [0,9]



histogram

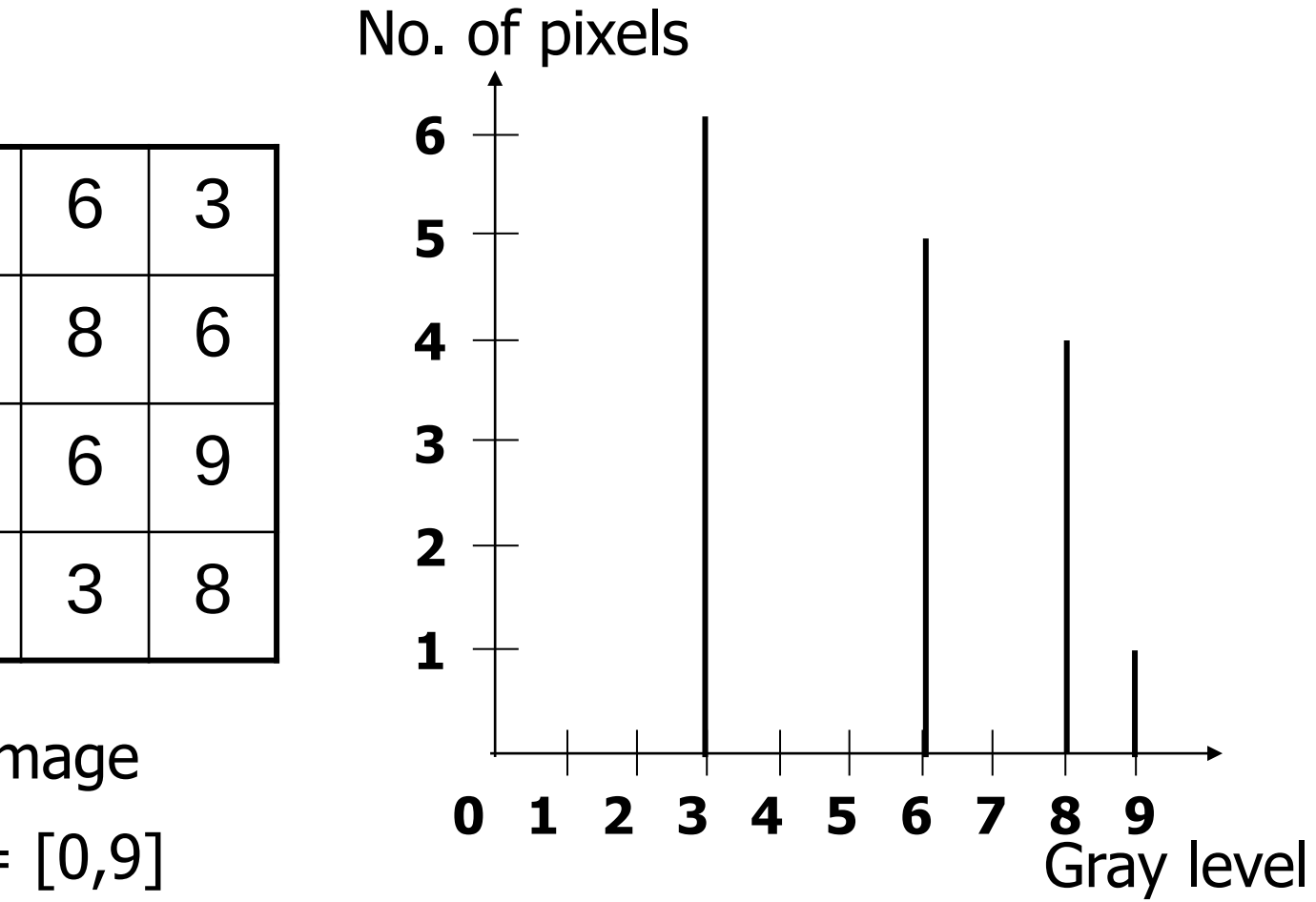
Gray Level(j)	0	1	2	3	4	5	6	7	8	9
No. of pixels	0	0	6	5	4	1	0	0	0	0
$\sum_{j=0}^{k} n_j$	0	0	6	11	15	16	16	16	16	16
$s = \sum_{j=0}^{k} \frac{n_j}{n}$	0	0	6 / 16	11 / 16	15 / 16	16 / 16	16 / 16	16 / 16	16 / 16	16 / 16
s x 9	0	0	3.3 ≈3	6.1 ≈6	8.4 ≈8	9	9	9	9	9

Contoh

3	6	6	3
8	3	8	6
6	3	6	9
3	8	3	8

Output image

Gray scale = [0,9]



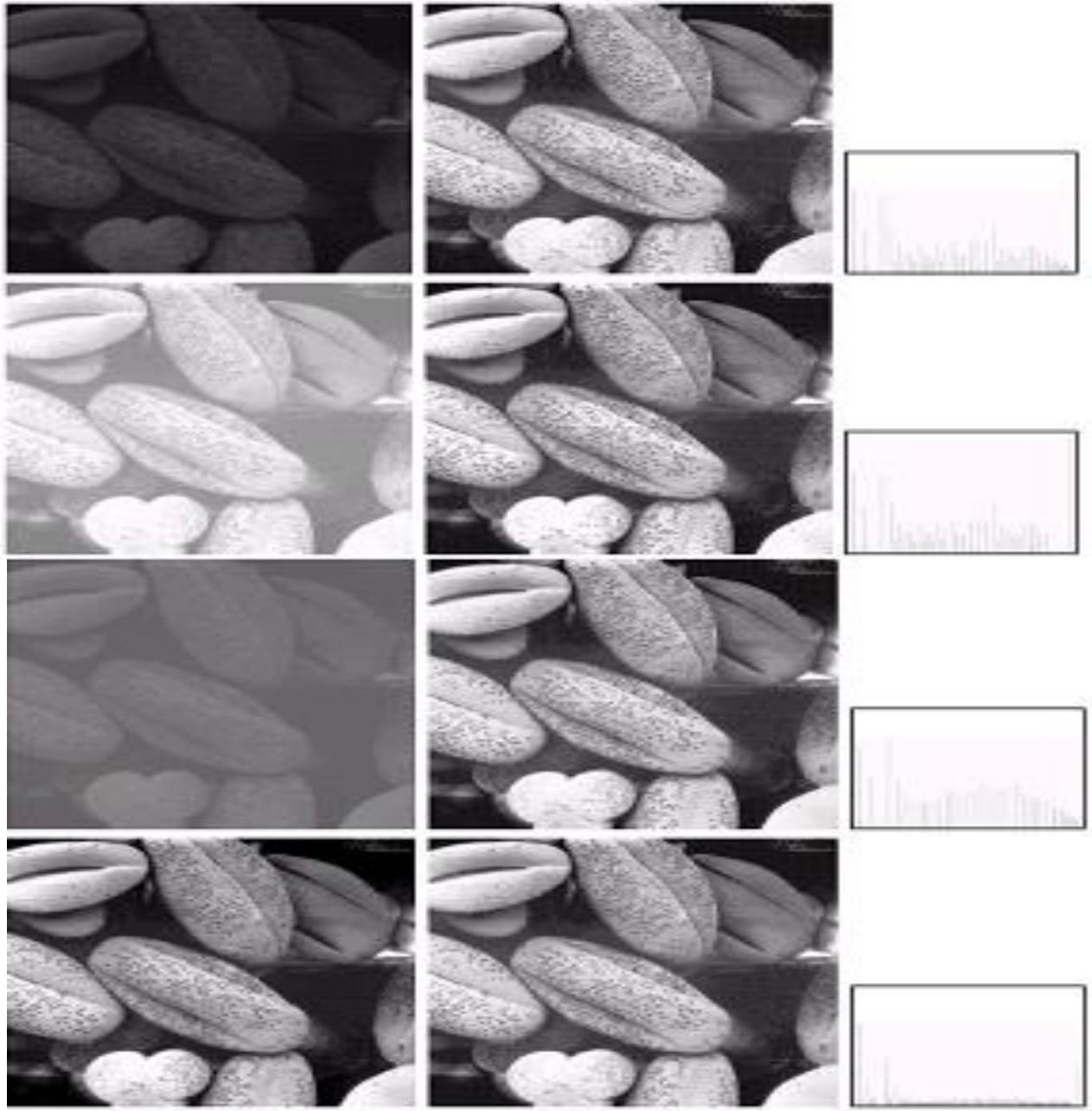
Histogram equalization

Program Matlab

```
function y=hist_eq(x)

[M,N]=size(x);
for i=1:256
h(i)=sum(sum(x==i-1));
end

y=x;s=sum(h);
for i=1:256
I=find(x==i-1);
y(I)=sum(h(1:i))/s*255;
end
```



- **Sebelum (kiri)**
- **Sesudah Histogram Equalization (kanan)**

Contoh

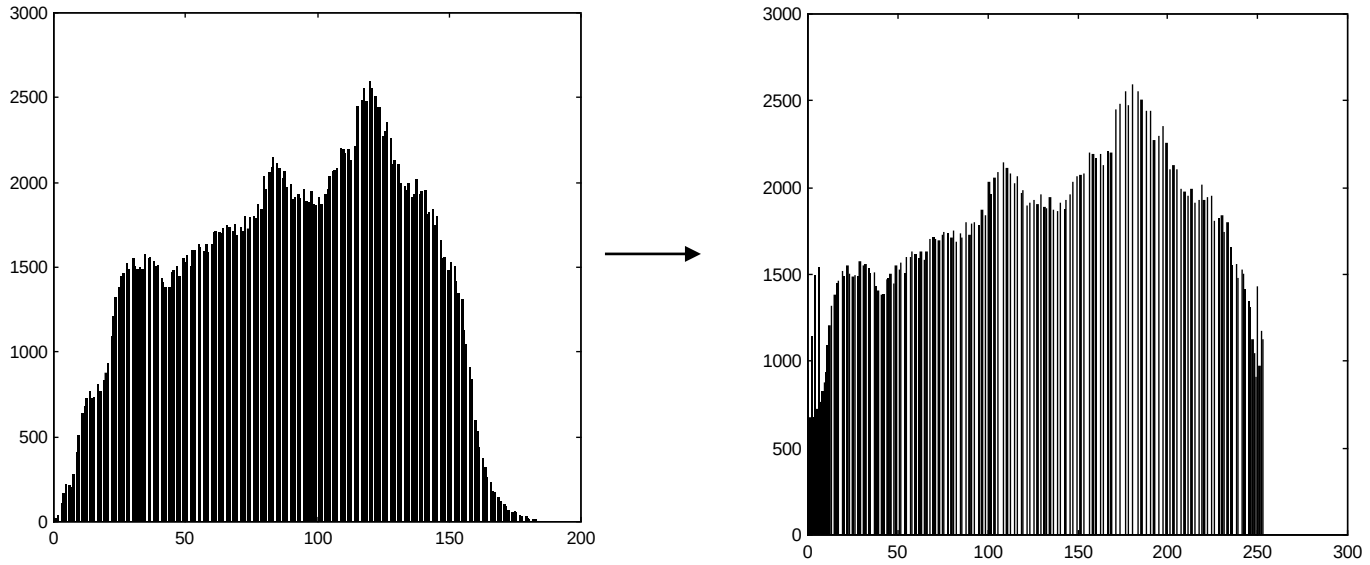


before



after

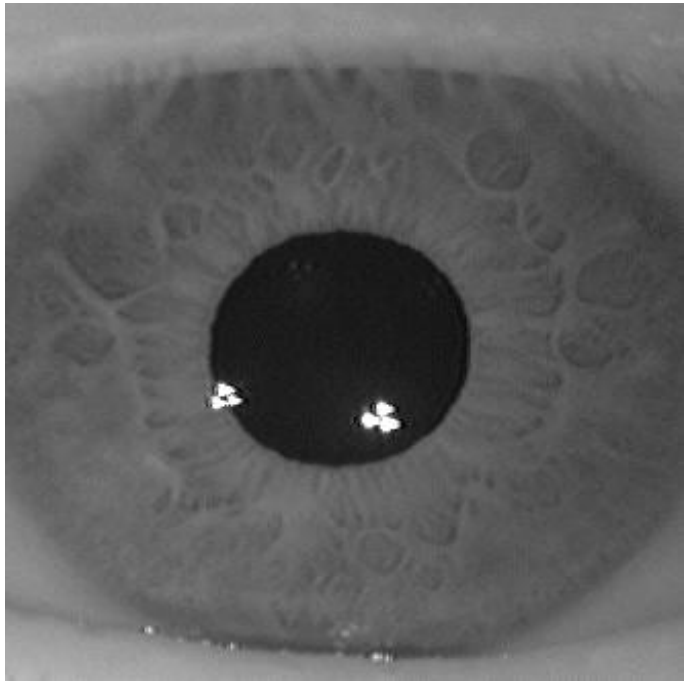
Perbandingan Histogram



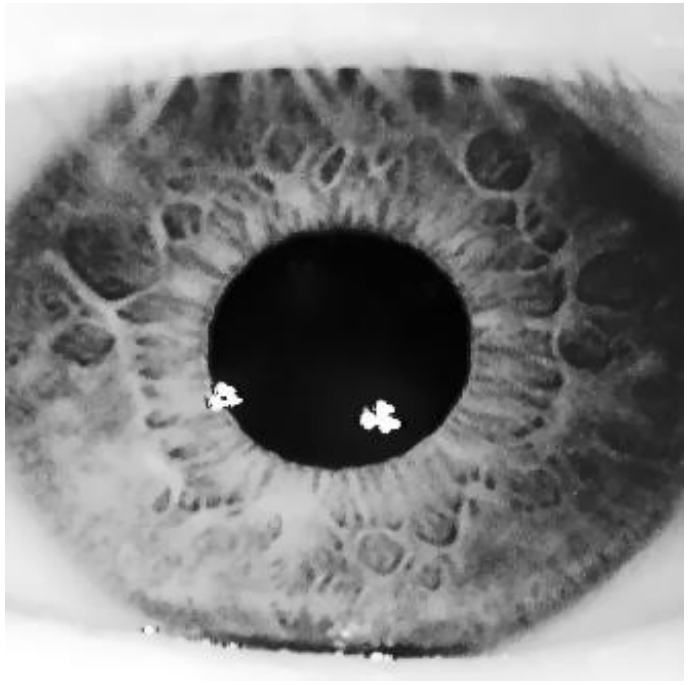
before equalization

after equalization

Pengenalan Iris



before

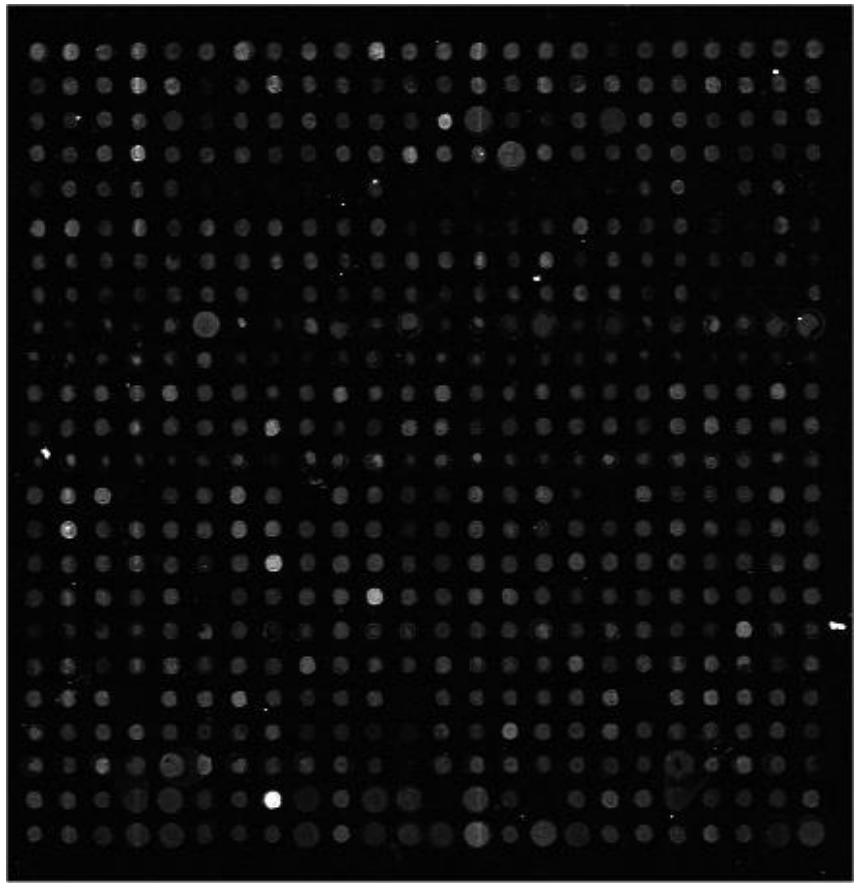


after

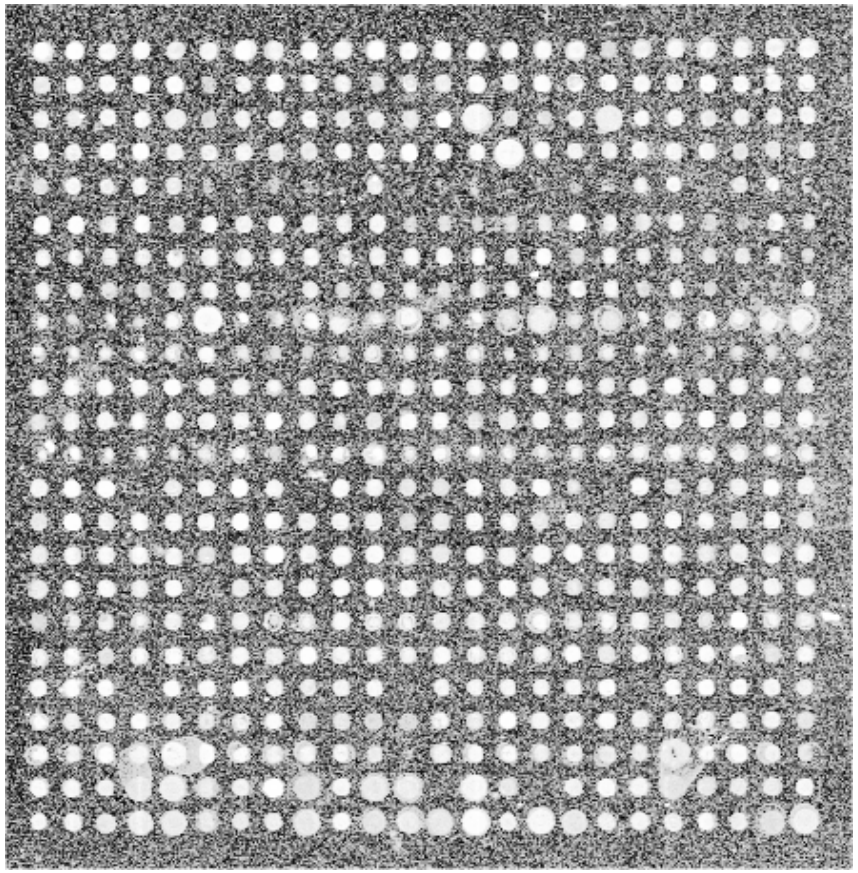
Image Bay



Microarray

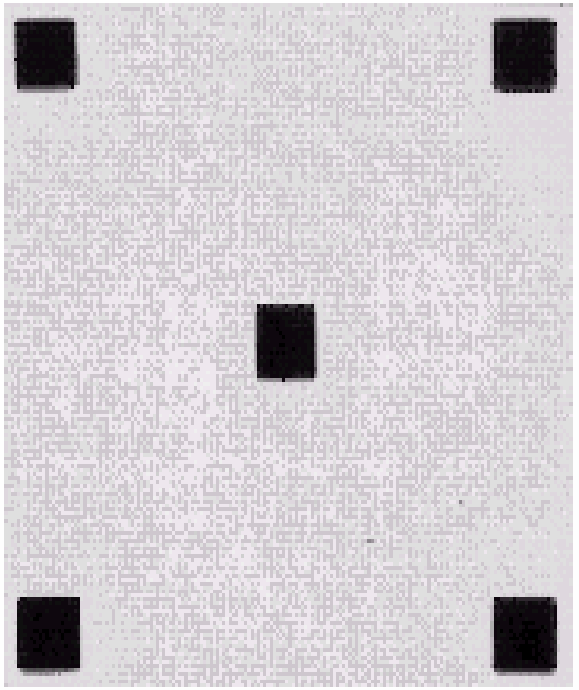


before

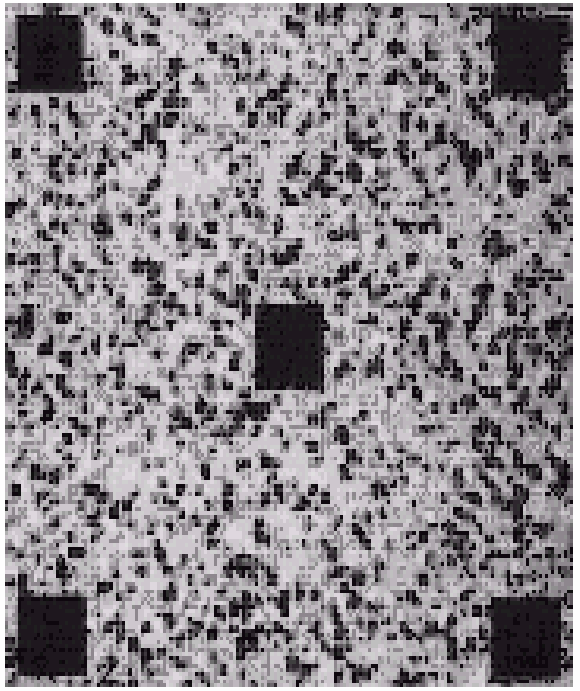


after

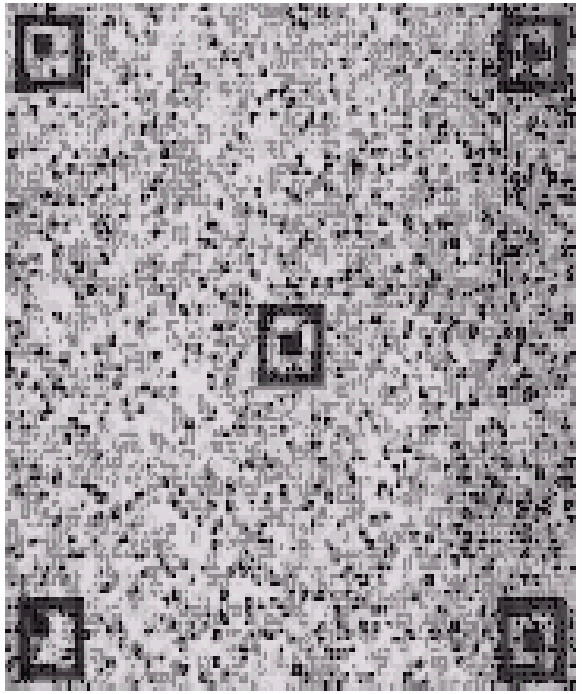
Contoh global dan lokal Histogram Equalization



Citra awal



Hasil global Histogram Equalization



Hasil local Histogram Equalization

Contrast Limited Adaptive Histogram Equalization (CLAHE)

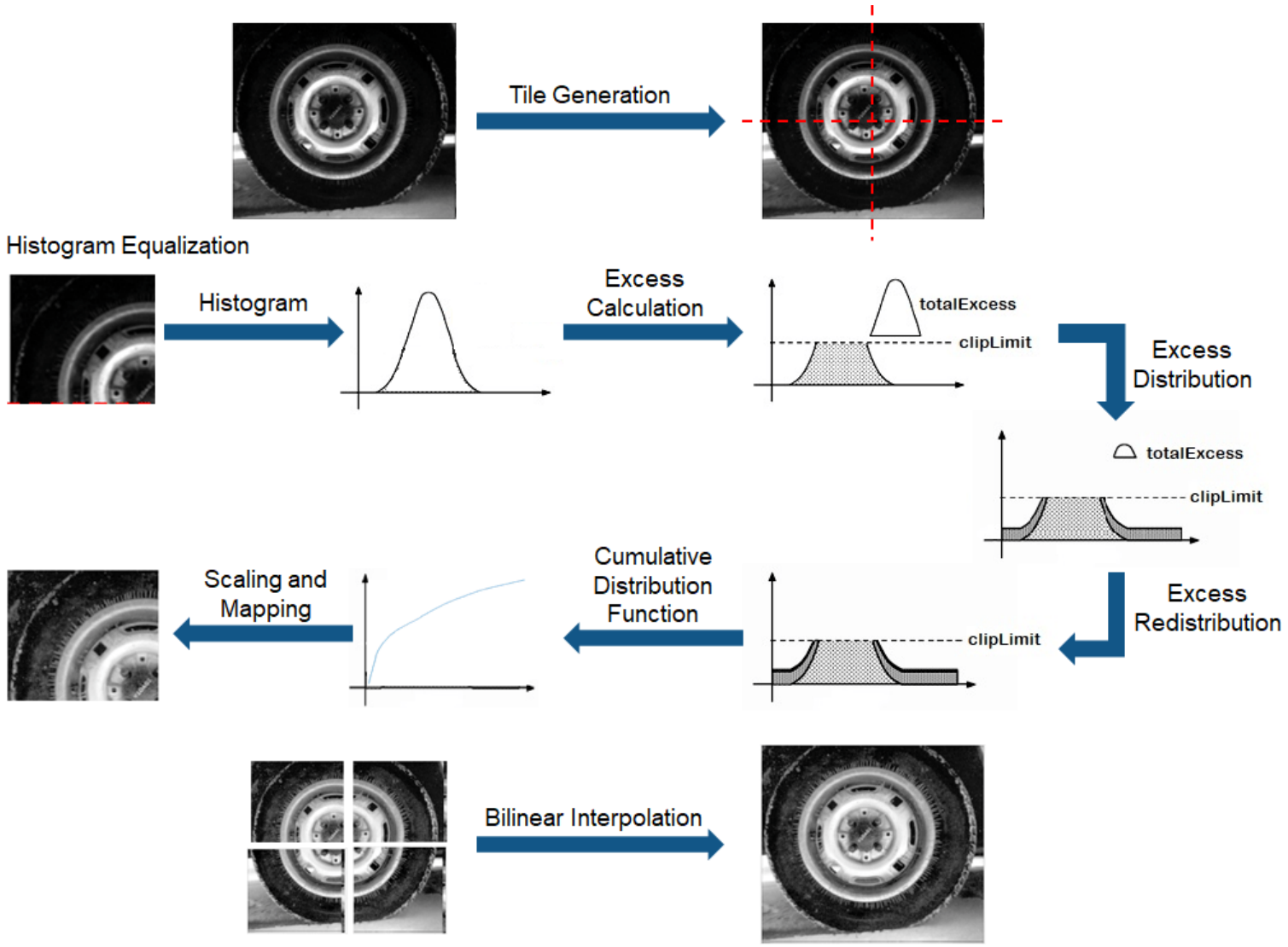
- *Adaptive histogram equalization* (AHE) adalah teknik praproses untuk memperbaiki kontras dalam citra, dengan menghitung beberapa histogram dari tile-tile (bagian citra) yang bersesuaian untuk meredistribusikan nilai luminan citra dan memperbaiki kontras local.
- CLAHE dikembangkan untuk mencegah amplifikasi noise yang berlebihan yang timbul dari *adaptive histogram equalization* (AHE) dengan membatasi amplifikasinya.
- CLAHE beroperasi pada area-area kecil (tile) dalam citra dengan histogram equalization (HE), nilai pada HE yang melebihi clipLimit dipotong dan total kelebihannya di redistribusikan kembali dengan parameter distribusi pada histogram.
- Tiap kontras tile ditingkatkan dengan penskalaan dan pemetaan dengan *cumulative distribution function* (CDF).
- Tile-tile yang bertetangga dikombinasikan dengan menggunakan interpolasi bilinier untuk memperhalus batas tepi tile.

Syntax:

J = adapthisteq(I)

J = adapthisteq(I,Name,Value)

Algoritma CLAHE (sumber: Matlab)



Contoh Program (sumber: Matlab)

```
clear all, close all;

% Read test image
I = imread('tire.tif');

% Enhance using global and adaptive histogram equalization
heqI = histeq(I);
adaptI1 = adapthisteq(I, 'NumTiles', [8 8]);
adaptI2 = adapthisteq(I, 'NumTiles', [16 16]);

% Show images
figure;
subplot(2, 2, 1), imshow(I), title('Input image');
subplot(2, 2, 2), imshow(heqI), title('histeq');
subplot(2, 2, 3), imshow(adaptI1), title('8*8 adapthisteq');
subplot(2, 2, 4), imshow(adaptI2), title('16*16 adapthisteq');
```

Hasil

Input image



histeq



8*8 adapthisteq



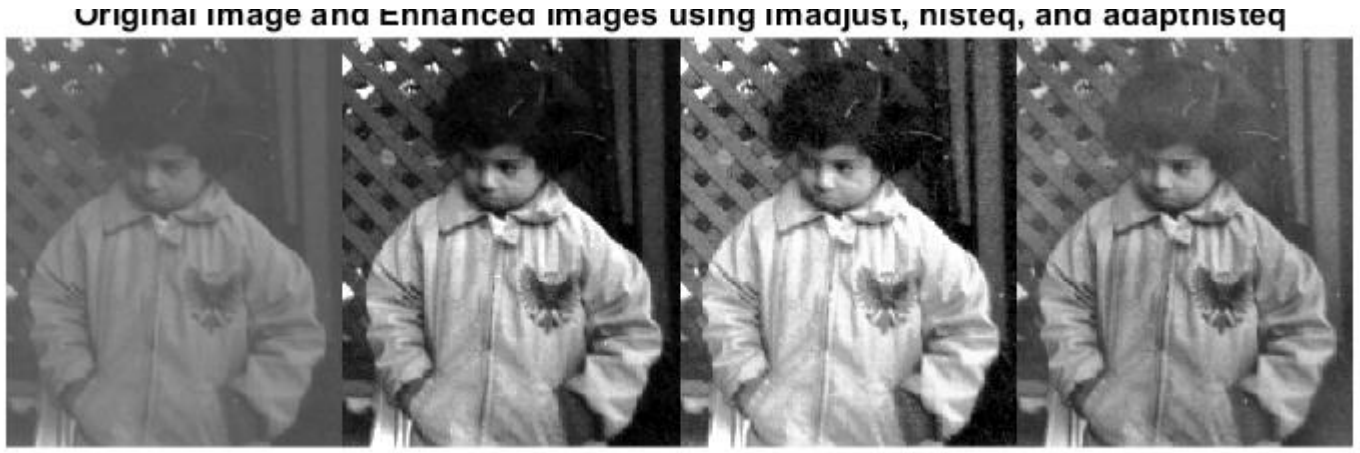
16*16 adapthisteq



Contoh Program (sumber: Matlab)

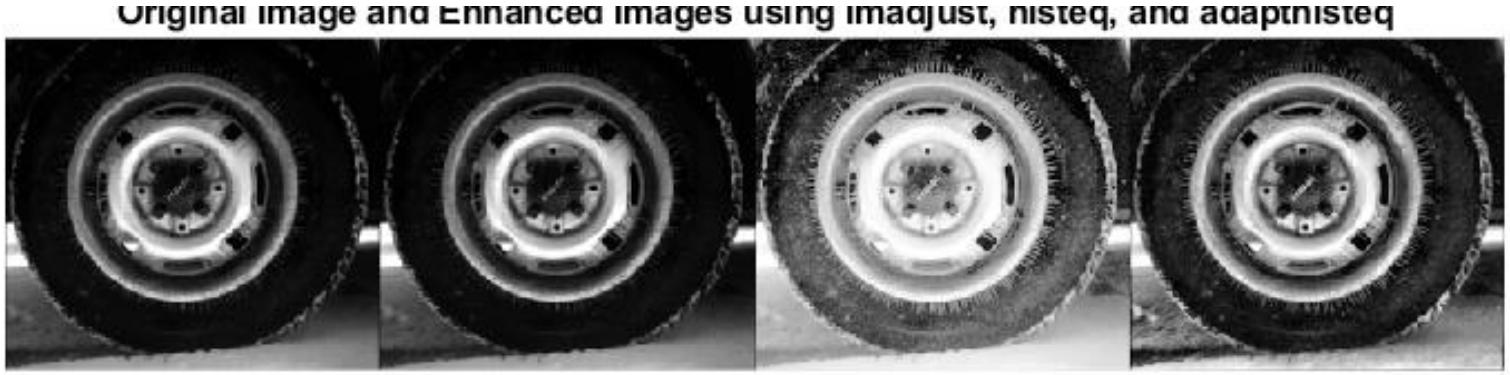
```
% Read a poor contrast image
pout = imread("pout.tif");
pout_imadjust = imadjust(pout);
pout_histeq = histeq(pout);
pout_adapthisteq = adapthisteq(pout);

% Display original image and three contrast adjusted images
montage({pout,pout_imadjust,pout_histeq,pout_adapthisteq},"Size",[1 4])
title("Original Image and Enhanced Images using imadjust, histeq, and
adapthisteq")
```

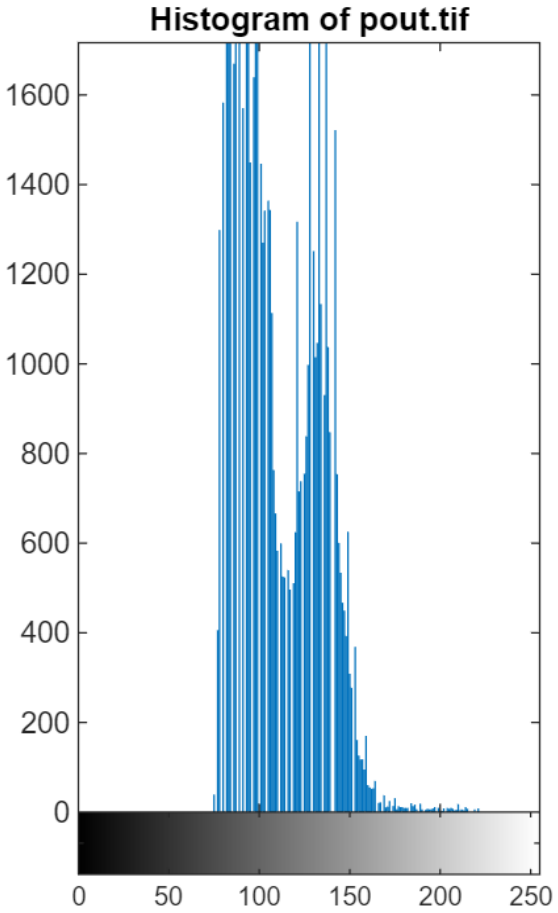


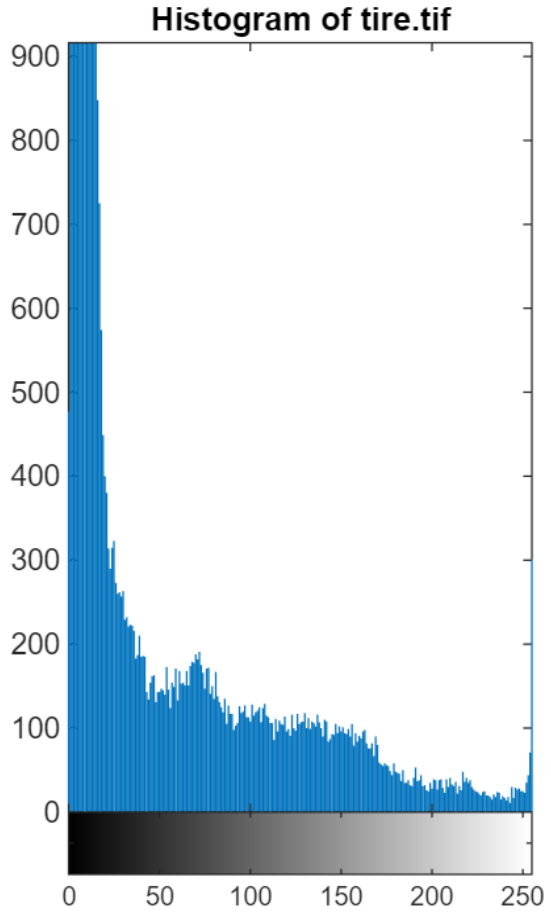
Contoh Program (sumber: Matlab)

```
tire = imread("tire.tif");
tire_imadjust = imadjust(tire);
tire_histeq = histeq(tire);
tire_adapthisteq = adapthisteq(tire);
```



Histogram dari citra beda kontras (sumber: Matlab)



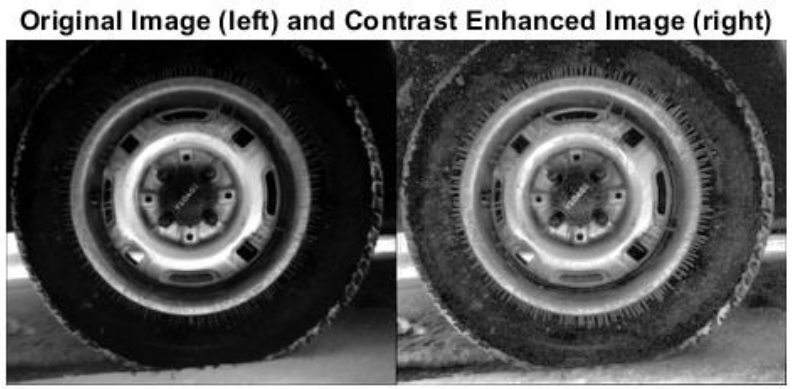



```
figure
subplot(1,2,1)
imhist(pout)
title("Histogram of pout.tif")
subplot(1,2,2)
imhist(tire)
title("Histogram of tire.tif");
```

Contoh Program (sumber: Matlab)

Syntax:
J = adapthisteq(I)
J = adapthisteq(I,Name,Value)

```
I = imread('tire.tif');
J = adapthisteq(I,'clipLimit',0.02,'Distribution','rayleigh');
imshowpair(I,J,'montage');
title('Original Image (left) and Contrast Enhanced Image (right)')
```



Spatial Filtering (Convolution)

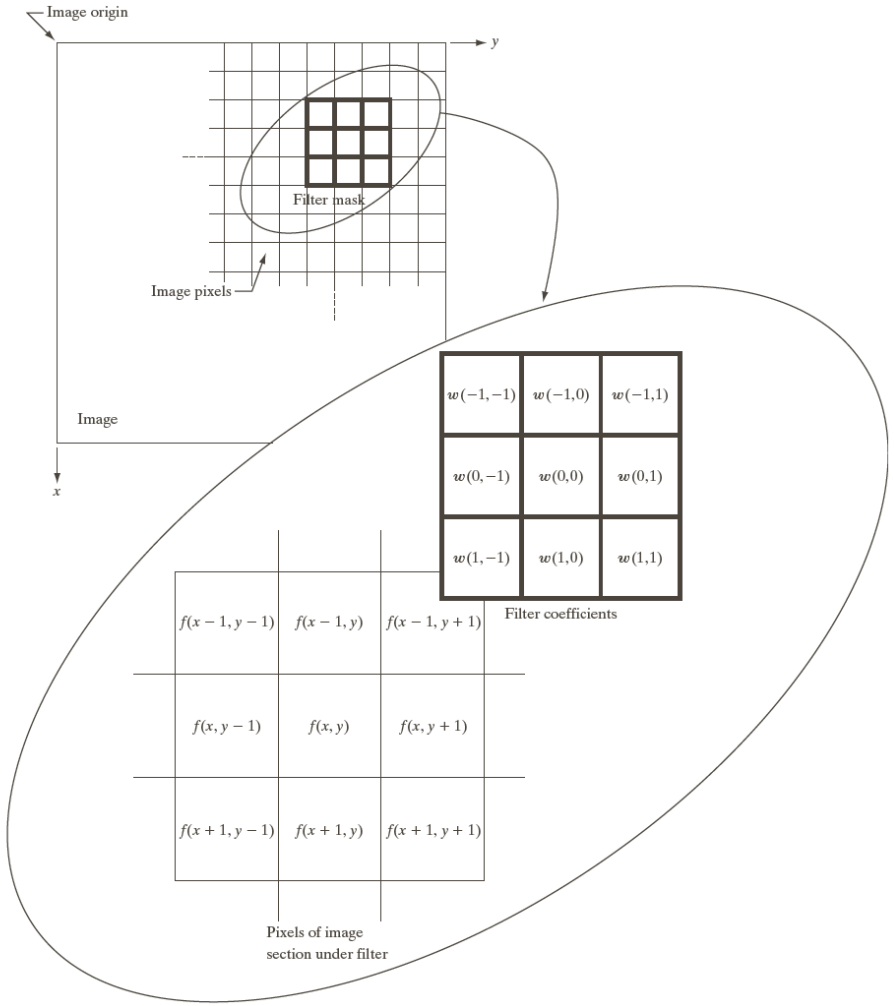


FIGURE 3.28 The mechanics of linear spatial filtering using a 3 × 3 filter mask. The form chosen to denote the coordinates of the filter mask coefficients simplifies writing expressions for linear filtering.

Low Pass Filter 3 x 3 (Smoothing, blurring)

$$\frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\frac{1}{16} \times \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

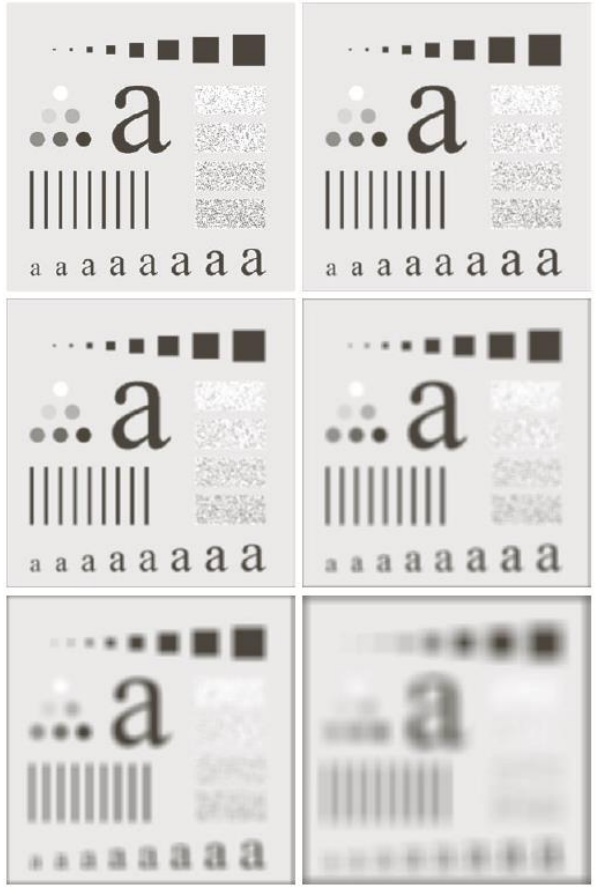
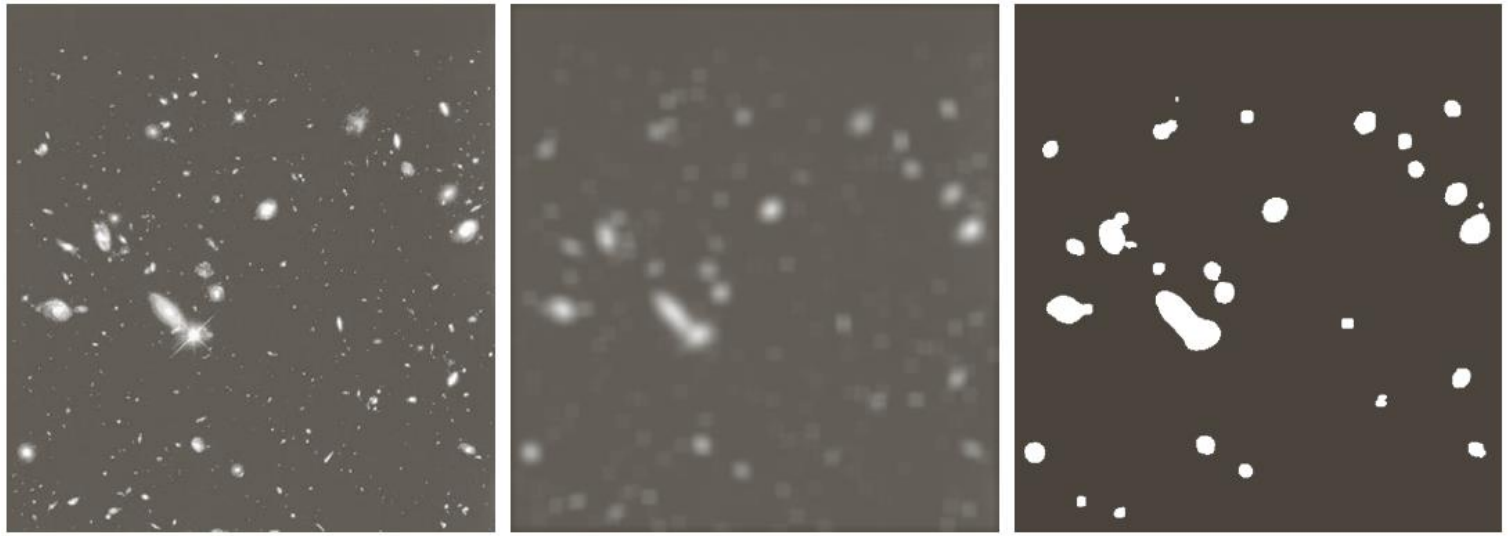


FIGURE 3.33 (a) Original image, of size 500 × 500 pixels. (b)–(f) Results of smoothing with square averaging filter masks of sizes $m = 3, 5, 9, 15$, and 35, respectively. The black squares at the top are of sizes 3, 5, 9, 15, 25, 35, 45, and 55 pixels, respectively; their borders are 25 pixels apart. The letters at the bottom range in size from 10 to 24 points, in increments of 2 points; the large letter at the top is 60 points. The vertical bars are 5 pixels wide and 100 pixels high; their separation is 20 pixels. The diameter of the circles is 25 pixels, and their borders are 15 pixels apart; their intensity levels range from 0% to 100% black in increments of 20%. The background of the image is 10% black. The noisy rectangles are of size 50 × 120 pixels.

a b
c d
e f



a b c

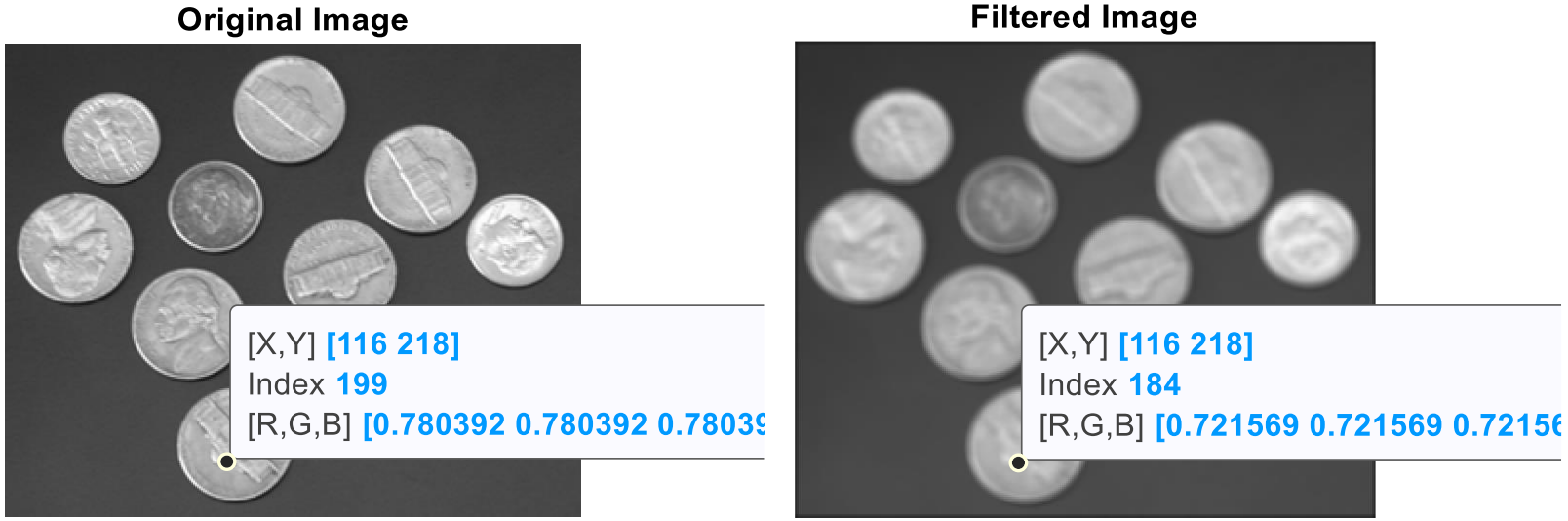
FIGURE 3.34 (a) Image of size 528 × 485 pixels from the Hubble Space Telescope. (b) Image filtered with a 15 × 15 averaging mask. (c) Result of thresholding (b). (Original image courtesy of NASA.)

Contoh Program (sumber: Matlab)

```
I = imread("coins.png");
figure; imshow(I), title("Original Image")

% 5-by-5, averaging filter.
h = ones(5,5)/25;

%Apply the averaging filter using imfilter.
I2 = imfilter(I,h);
figure; imshow(I2), title("Filtered Image")
```



Citra grayscale dengan rentang nilai 0 - 1

Contoh Program (sumber: Matlab)

```
rgb = imread("peppers.png");
figure; imshow(rgb);
h = ones(5,5)/25;
rgb2 = imfilter(rgb,h);
figure; imshow(rgb2)
```





Laplacian Masks

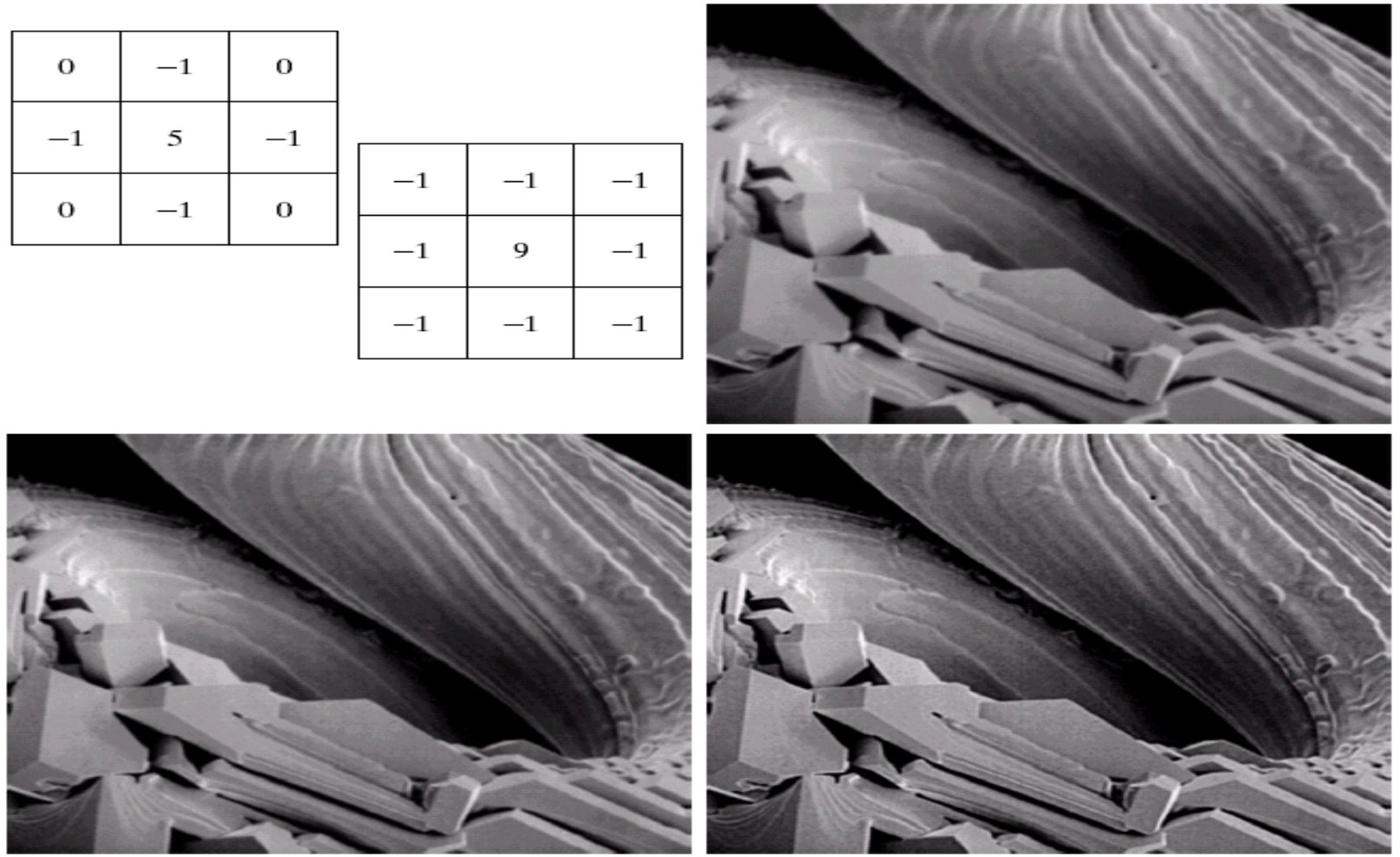
0	−1	0
−1	4	−1
0	−1	0

−1	−1	−1
−1	8	−1
−1	−1	−1

High Pass Filter 3 x 3 (HPF)

0	−1	0
−1	5	−1
0	−1	0

−1	−1	−1
−1	9	−1
−1	−1	−1



a b c
d e

FIGURE 3.41 (a) Composite Laplacian mask. (b) A second composite mask. (c) Scanning electron microscope image. (d) and (e) Results of filtering with the masks in (a) and (b), respectively. Note how much sharper (e) is than (d). (Original image courtesy of Mr. Michael Shaffer, Department of Geological Sciences, University of Oregon, Eugene.)

Operator 2x2 Roberts cross-gradient

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

$$G_x = (z_9 - z_5) \quad \text{and} \quad G_y = (z_8 - z_6)$$

$$\nabla f = [G_x^2 + G_y^2]^{1/2} = [(z_9 - z_5)^2 + (z_8 - z_6)^2]^{1/2}$$

$$\nabla f \approx |z_9 - z_5| + |z_8 - z_6| \rightarrow$$

-1	0
0	1

0	-1
1	0

Operator Sobel 3x3

$$G_x = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)$$

$$G_y = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$$

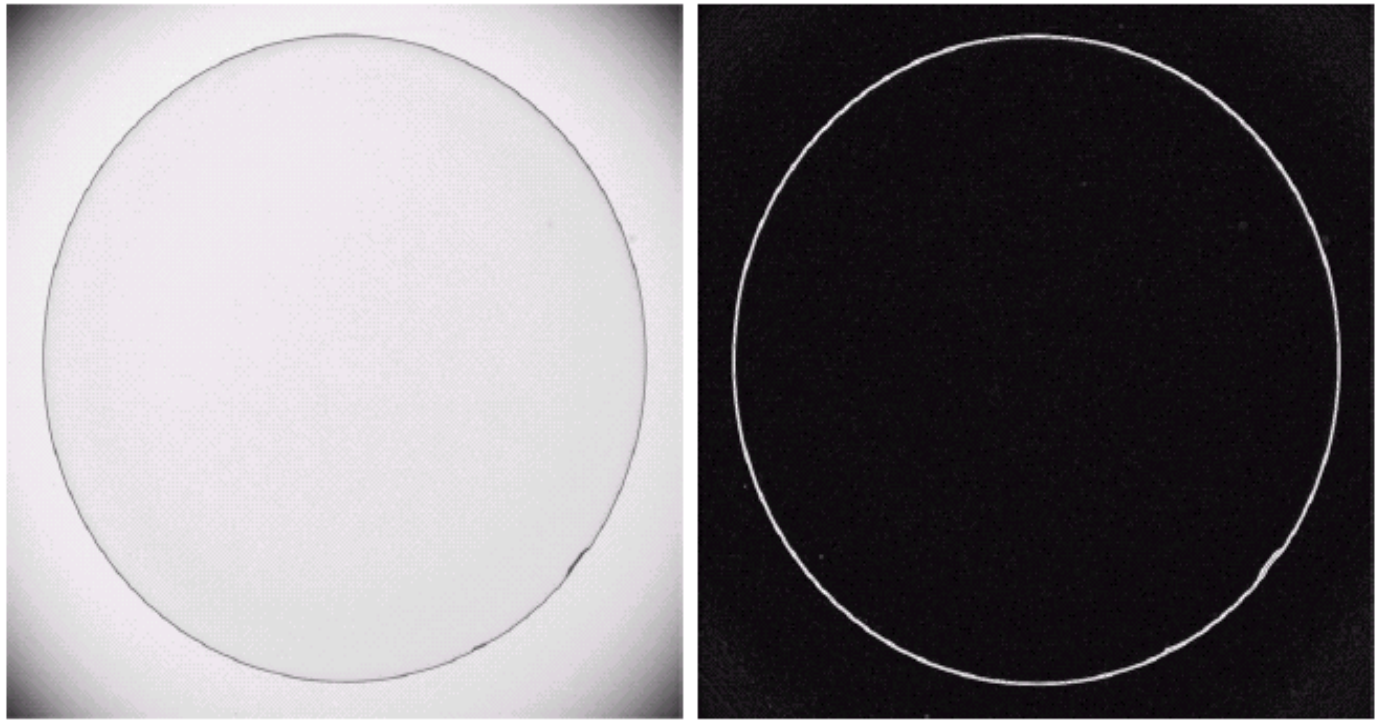
z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

−1	−2	−1
0	0	0
1	2	1

−1	0	1
−2	0	2
−1	0	1

$$\nabla f \approx |G_x| + |G_y|$$

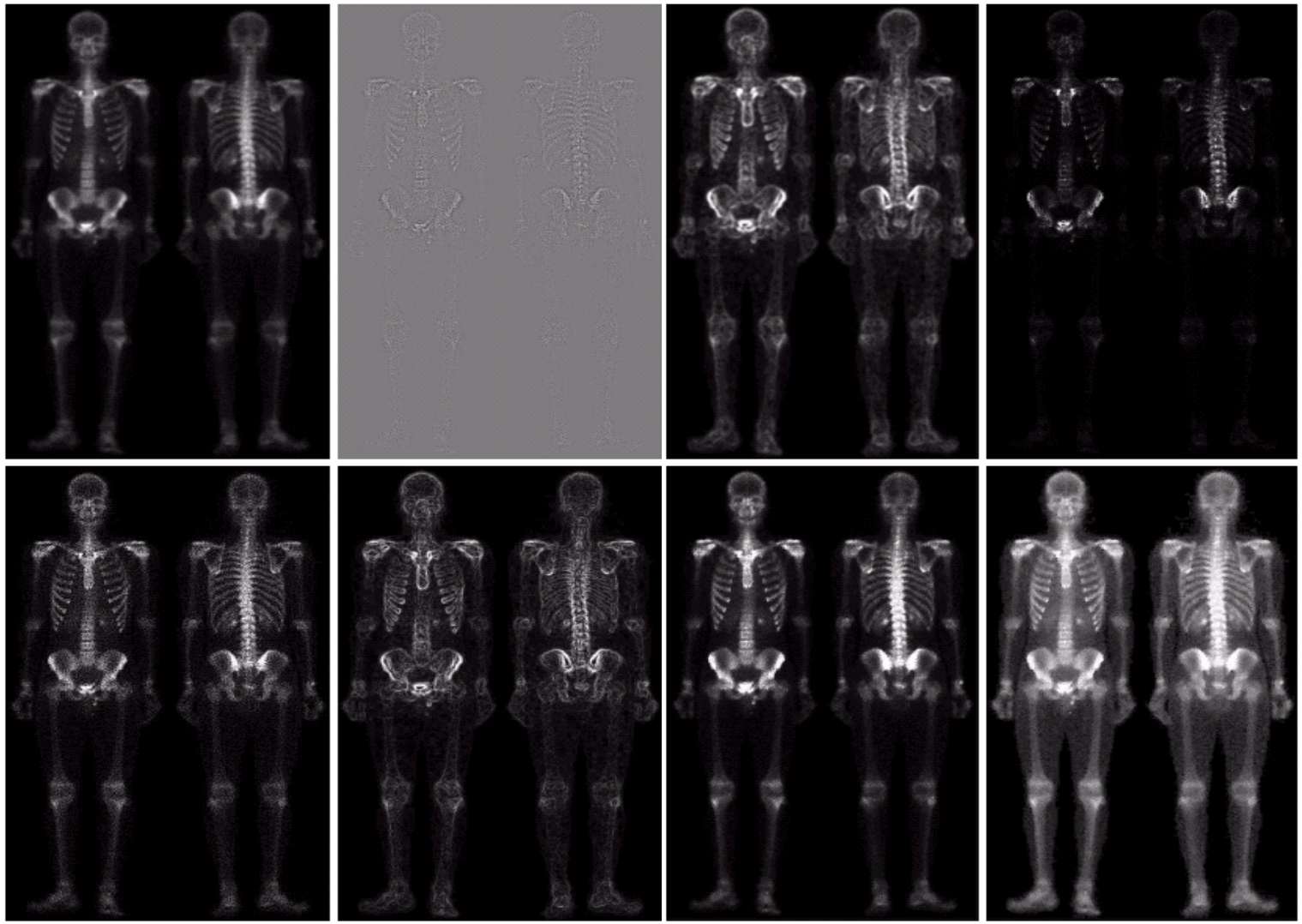
Contoh Gradient Sobel



a b

FIGURE 3.45
Optical image of contact lens (note defects on the boundary at 4 and 5 o'clock). (b) Sobel gradient. (Original image courtesy of Mr. Pete Sites, Perceptics Corporation.)

Contoh Laplacian dan Sobel



a b
c d

FIGURE 3.46
(a) Image of whole body bone scan.
(b) Laplacian of (a). (c) Sharpened image obtained by adding (a) and (b). (d) Sobel of (a).

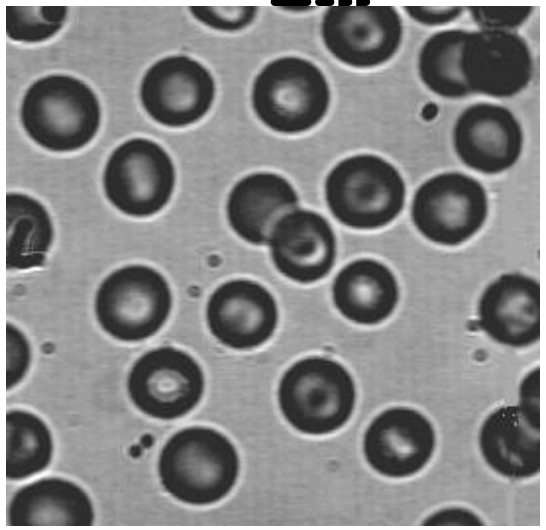
e f
g h

FIGURE 3.46
(Continued)
(e) Sobel image smoothed with a 5×5 averaging filter. (f) Mask image formed by the product of (c) and (e).
(g) Sharpened image obtained by the sum of (a) and (f). (h) Final result obtained by applying a power-law transformation to (g). Compare (g) and (h) with (a). (Original image courtesy of G.E. Medical Systems.)

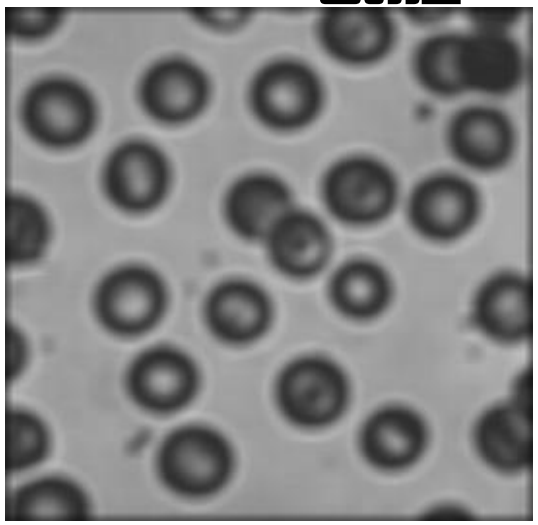
Contoh Matlab LPF

```
>> im = imread('filename');
>> imshow(im)
>> h = fspecial('average', 9);
>> im2 = uint8( round( filter2(h, im)));
>> imshow(im2)
```

im

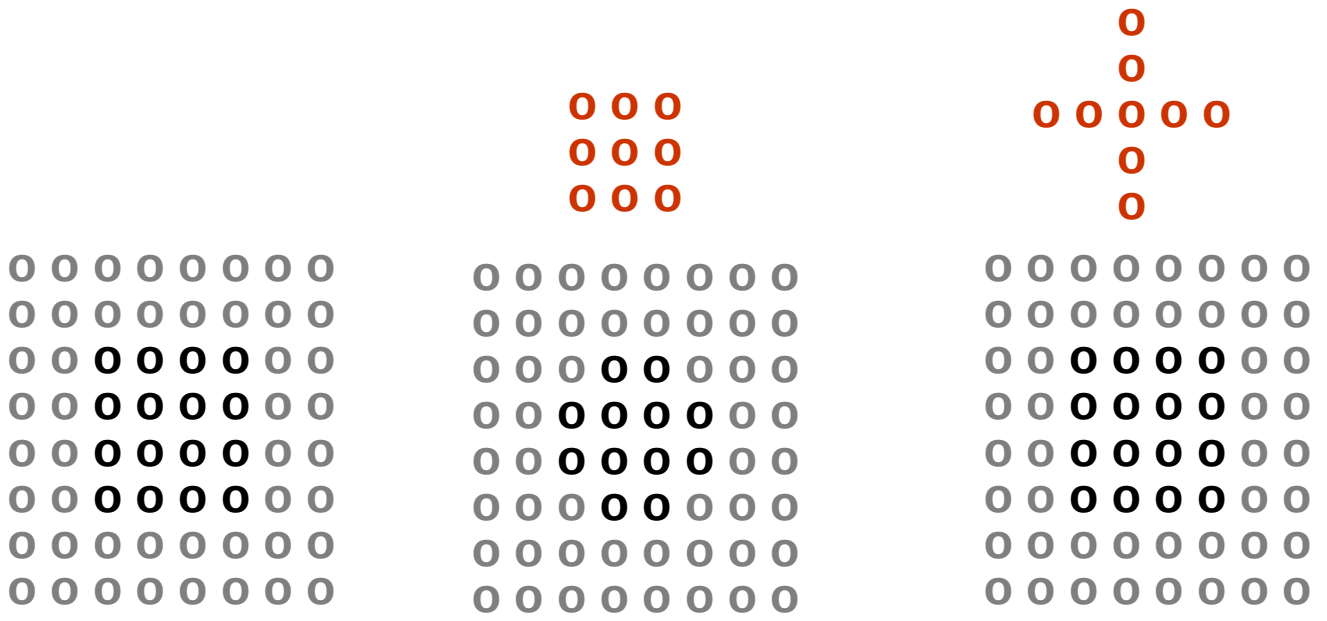


im2

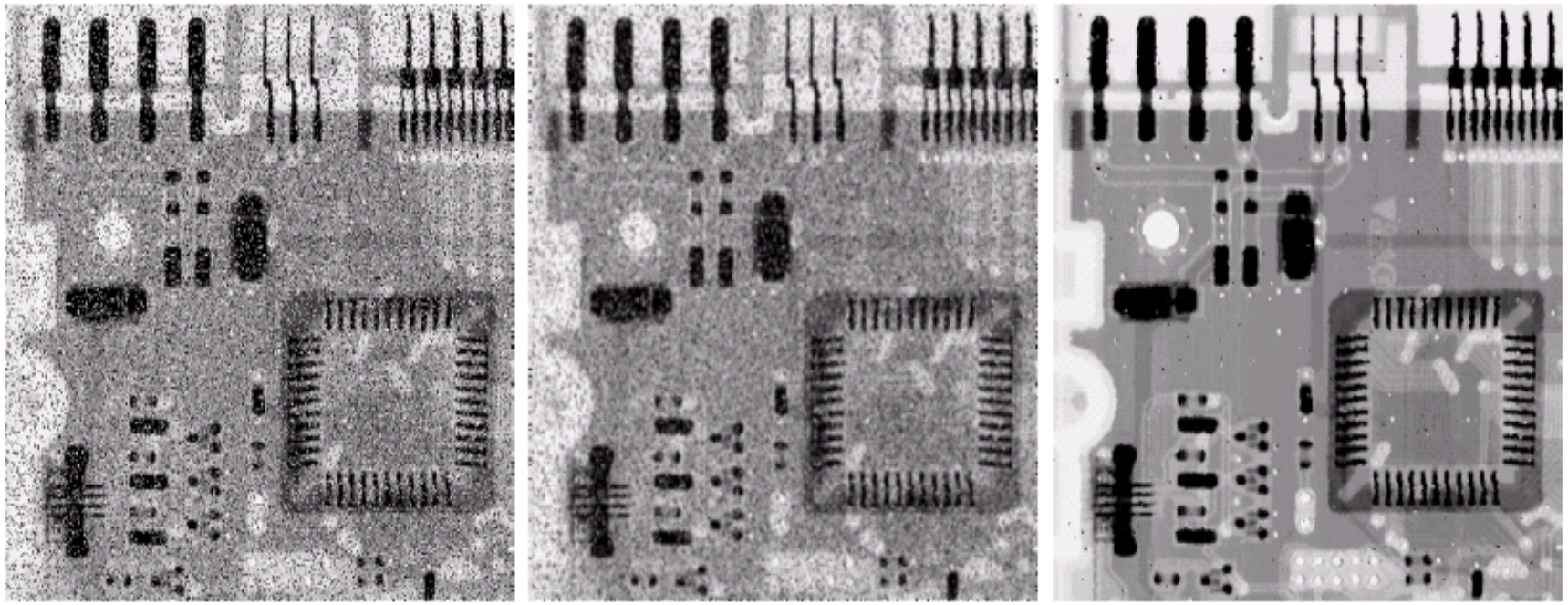


Filter Median

- Mengganti harga piksel dengan median dari piksel dan tetangganya of the median)
- Reduksi random noise dengan hasil kurang kabur dibandingkan filter penghalus (smoothing)
- Eliminasi piksel2 terisolasi



Contoh Filter Median



a b c

FIGURE 3.37 (a) X-ray image of circuit board corrupted by salt-and-pepper noise. (b) Noise reduction with a 3×3 averaging mask. (c) Noise reduction with a 3×3 median filter. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)


THANK YOU